

Lie-Group FullVA Integration for Cylindrical Lower-Pair Mechanisms: A Conditional Sixth-Order Formal-Order Comparison with Separated Four-Example Evidence

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v047 artifact-backed paper build

Abstract

This paper reports the v047 evidence for a Lie-group FullVA integrator applied to cylindrical lower-pair mechanisms. The accepted Gauss6/FullVA path is evaluated on the cylindrical-chain benchmark and on four ASME-style mechanism examples with separated evidence roles: single pendulum, double pendulum, four link, and slider crank. On the smooth cylindrical h-sweep, the method records observed position and velocity orders 7.161 and 7.066. The sharp Brown–McPhee friction case is order-reduced on the coarse h-sweep, but high-order behavior reappears under deeper fixed refinement, leaving a practical coarse-regime caveat rather than a closed-form correctness failure.

The main claim of this paper is therefore a conditional formal-order comparison: within the current artifact scope, the accepted Gauss6/FullVA path has a conditional sixth-order Lie-group FullVA theorem on the selected smooth branch, while the four examples provide separated method-side evidence roles. The single- and double-pendulum rows support dynamic-order evidence; the four-link and slider-crank rows support lower-pair mechanism coverage. The local paper-style $m = 3$ Lobatto-TFE formula target is order five. A full-TFE replacement would strengthen the reproduction claim, but it is not needed for this bounded formal-order comparison.

The paper also separates this accepted result from a stricter reproduction claim. The original Chaturvedi–Sandu–Sandu paper has its own full TFE formulation. In this v047 implementation, “full-TFE replacement” means replacing the accepted Gauss6/FullVA stage residual by paper-derived temporal-finite-element weak rows inside the nonlinear solve. That replacement gate remains open and machine-readable as `full_tfe_stage_replacement=false`. The open replacement gate is reported as a limitation and as future reproduction work; it does not weaken the accepted conditional method/proof claim or the separated four-example evidence roles.

Reviewer-Facing Summary

The current paper should be read as an artifact-backed status paper, not as a claim that every research gate is closed. The accepted method claim is the conditional sixth-order Gauss6/FullVA path under the direct residual-bridge proof route. The cylindrical-chain benchmark records smooth position/velocity orders 7.161/7.066. The four ASME-style examples have separated roles: single and double pendula provide dynamic-order evidence, while four link and slider crank provide mechanism-coverage and reaction/constraint diagnostics.

The paper-TFE claim is separate. The encoded paper formula map uses $m = 3$ Gauss–Lobatto temporal finite elements and has expected order five in the formula-mapping audit, but this is

not yet an accepted method replacement. The machine-readable gate remains `full_tfe_stage_replacement=false`. The remaining missing object is an eight-row source-free lower-pair closure that replaces the terminal bridge inside the 132-row nonlinear stage residual while preserving the paper-TFE order target. Equivalently, the source paper is not missing a TFE method; the missing piece is our artifact-backed substitution of that paper-style residual into the v047 solver. The newest endpoint-pose trajectory smoke has narrowed that missing object: the terminal-equivalent endpoint row closes velocity but loses order, while a nearby nonterminal row preserves neither terminal closure nor order. The next repair therefore has to be a nonterminal endpoint-pose predictor that closes terminal velocity on trajectory without collapsing the h-sweep order. Sliding the extrapolation points almost onto the endpoint closes terminal velocity, but it reproduces the same 4.142/2.305 smooth-order loss; quadratic nonterminal extrapolation does the same.

Routine paper validation should use the read-only validators, not the full generator. The historical `run_v047.py` file is an accumulated audit harness containing positive evidence, negative screens, and artifact generation code; the latest recorded full generation time is 2386.76 s. It is not needed to check the accepted four-example claim or rebuild this PDF.

1 Scope and Claims

The v047 pipeline has one primary paper goal and one optional stronger reproduction goal. The primary goal is to validate a conditional sixth-order Lie-group multibody integrator for cylindrical lower-pair mechanisms. This is the accepted Gauss6/FullVA path: it has a conditional sixth-order theorem on the selected smooth branch by the direct 132-row residual bridge, records smooth position/velocity orders 7.161/7.066 in the current h-sweep, and has four-example coverage with separated evidence roles. The single- and double-pendulum rows provide dynamic-order evidence, while the four-link and slider-crank rows provide closed-loop constraint/reaction checks that are not interpreted as accepted asymptotic order estimates. The optional stronger goal audits whether a complete paper-derived temporal finite-element stage residual can replace the current stage equations without relying on endpoint projection, endpoint source data, or terminal-row substitution. This second goal is intentionally stricter and is not needed for the formal-order comparison claim.

The source paper itself is therefore not being accused of lacking a full TFE formulation. The open v047 item is narrower and implementation-facing: we have not yet produced an accepted nonlinear solve in which the paper-style TFE stage rows replace the Gauss6/FullVA stage rows while retaining the required order, rank, residual, and endpoint-closure evidence.

The full-TFE replacement is therefore not needed to state the four-example method evidence or the bounded formal-order comparison. It is needed only for the stronger statement that the implemented nonlinear residual is the original paper-style temporal finite-element method rather than the accepted conditional Gauss6/FullVA method proved through the direct 132-row residual bridge with separate diagnostic closure evidence. Keeping this boundary explicit is important because otherwise the formal-order comparison could be confused with a solved paper-method reproduction.

The distinction matters for interpreting the numerical order. The accepted Gauss6/FullVA path is high order on smooth data. The current source-free final-stage TFE closure candidate and the newer endpoint-pose velocity predictor smoke are not accepted replacements: they close or nearly close terminal velocity, but they do not preserve the target trajectory order. The paper therefore reports the Gauss6/FullVA theorem and method evidence as the positive result and the full TFE replacement as an open proof and implementation gate.

Table 1: Terms used in the v047 paper and pipeline.

Term	Meaning in this paper
Gauss6/FullVA	The accepted Gauss collocation/FullVA trajectory residual: a conditional order-six method path on the selected smooth branch by the direct 132-row residual bridge.
Paper-TFE	The temporal finite-element formula family encoded from the paper coefficient maps; the v047 formula-mapping audit uses $m = 3$ Gauss–Lobatto nodes with expected order five.
Independent full-TFE replacement	All 132 stage rows are replaced by paper-derived stage rows and are solved inside Newton, with accepted h-sweep order, residual, rank, and endpoint checks.
Source-free lower-pair closure	The lower-pair closure rows do not receive an external endpoint source and do not rely on endpoint-boundary source data.
Terminal-row replacement	A diagnostic shortcut that inserts raw endpoint velocity rows in place of paper rows. This is useful for debugging but is not accepted as full TFE.
Projection	A post-step velocity correction. Projection is accepted only on the historical Gauss diagnostic path, not on the independent full-TFE replacement path.

1.1 Terminology and Claim Boundary

The phrase *independent full-TFE stage replacement* is used in a narrow technical sense. It does not mean merely adding an endpoint correction to an otherwise accepted Gauss trajectory. It means replacing the stage residual rows themselves by paper-derived temporal finite-element rows and solving those rows inside the nonlinear trajectory problem.

Thus the current open issue is not whether the four examples run. They do. It is whether the stricter paper-TFE residual can replace the Gauss stage equations without losing the order and endpoint behavior already demonstrated by the accepted Gauss6/FullVA path. The occasional spelling “FTE” in notes should be read as TFE, temporal finite element.

1.2 Claim Map

The current result has one primary accepted formal-order comparison, one accepted four-example method gate, and one optional open reproduction claim. The primary claim is that Gauss6/FullVA is a conditional sixth-order method in the artifact scope used here: it has observed smooth orders 7.161/7.066, whereas the encoded local paper-style $m = 3$ Lobatto-TFE target has expected order five. The open claim is narrower: a paper-derived full-TFE stage residual has not yet independently replaced the Gauss stage equations.

1.3 Comparator Definition

The comparison in Proposition 1 is deliberately local and reproducible. The comparator is the paper-style $m = 3$ Gauss–Lobatto TFE formula target encoded in the v047 formula-mapping audit, not an independently accepted implementation of every source-paper residual row. This keeps the accepted claim aligned with the available evidence: the pipeline can prove that the conditional

Gauss6/FullVA path has formal order six while the local paper-style target has expected order five, while still reporting that the complete source-paper residual reproduction remains open.

1.4 Main Comparative Proposition

Proposition 1 (Formal-order comparison). *Within the current v047 artifact scope, the accepted Gauss6/FullVA path supports a conditional formal-order comparison relative to the encoded local paper-style $m = 3$ Lobatto-TFE target.*

Evidence and proof sketch. The accepted method path is the Gauss/FullVA residual on the cylindrical lower-pair benchmark. Its order statement is the conditional theorem proved by the direct 132-row residual bridge: on the selected smooth compact branch, the 96 non-dynamic rows supply row-local $O(h^7)$ residual input, the 36 Newton–Euler rows vanish by direct substitution on the lifted Gauss stage, and endpoint/Newton perturbations are retained at $O(h^7)$ before the local-to-global $O(h^6)$ transfer. The smooth h-sweep records observed position and velocity orders 7.161/7.066 as downstream numerical evidence, and the same pipeline accepts the four-example evidence gate. Only the single- and double-pendulum dynamic rows are used as method-order evidence; the four-link and slider-crank rows remain closed-loop constraint/reaction checks whose near-roundoff trajectory differences are not interpreted as convergence order. The comparator used for this proposition is not a complete source-paper residual implementation; it is the locally encoded $m = 3$ Gauss–Lobatto TFE formula target, whose expected order is five in the formula-mapping audit. Therefore the artifact-backed comparison is between a conditional order-six Gauss/FullVA method path and a local order-five paper-style target. This proves the bounded paper-facing comparative statement claimed here.

The proposition is intentionally not a full residual reproduction theorem. A full source-paper reproduction statement would require the still-open independent full-TFE stage replacement gate, where the paper-derived rows replace the Gauss6/FullVA stage residual inside the nonlinear solve and pass the rank, residual, endpoint-closure, and h-sweep checks without endpoint source data, terminal-row replacement, projection, or a target-direction oracle. That stronger statement remains outside the accepted formal-order comparison. $\square$

1.5 Contributions

The paper is organized around artifact-backed claims rather than a single final theorem. The current contributions are:

- a conditional order-six integrator comparison for the cylindrical lower-pair benchmark: the accepted Gauss6/FullVA path is proved through the direct 132-row residual bridge and records smooth position and velocity orders 7.161/7.066, while the local paper-style $m = 3$ Lobatto-TFE formula target has expected order five;
- accepted method-side evidence for four ASME-style examples, including lower-pair graph, rank, closed-loop kinematic FullVA, and reaction-dynamics checks for the closed-loop examples;
- a paper-TFE formula-mapping and substitution audit that keeps the expected fifth-order Lobatto-TFE claim distinct from the accepted Gauss6 method claim;
- a source-free final-stage lower-pair closure that removes endpoint source data, terminal-row replacement, and projection while closing raw terminal velocity to roundoff;

- a negative but useful compression-audit sequence showing which source-free eight-row closure families do not yet provide an independent full-TFE replacement.

1.6 Role of the Full-TFE Replacement

The full-TFE replacement is not needed to support the accepted Gauss6 result: the order-six claim is conditional on the retained compact-branch, endpoint, binding, and solver-scale interfaces, with smooth numerical evidence downstream. The replacement is optional future work for a stronger source-paper reproduction claim. If the paper is to claim a temporal finite-element stage method rather than a Gauss-stage method with endpoint diagnostics, then the rows solved by Newton must be the paper-derived weak/TFE rows themselves. In particular, the lower-pair endpoint closure must be produced by stage-local source-free rows, not by an endpoint source, a final-row selector, a projection step, or a local target-Jacobian oracle.

This distinction is why the pipeline keeps running targeted compression audits after the four-example method gate is accepted. Those audits are not redoing the four examples. They are isolating the specific eight-row lower-pair closure law that would make the paper-TFE residual independently viable.

1.7 Reader Status Checklist

The current status can be read from four separate questions. This separation prevents the accepted method evidence from being confused with the still-open paper-TFE replacement evidence.

2 Numerical Model

The v047 cylindrical-chain benchmark replaces twist-locked prismatic joints by cylindrical joints. The cylindrical lower pair constrains axis alignment and a perpendicular point relation while leaving axial translation and relative spin free. The main residual has dimension 132. Its sparse Jacobian pattern has 2637 nonzeros; column coloring uses 90 colors, and the row-VJP backend uses 60 row colors while preserving trajectory agreement to roundoff.

The accepted trajectory integrator uses a three-stage Gauss collocation structure with FullVA variables and is claimed through the conditional order-six residual-bridge theorem. Each body carries position, orientation, velocity, angular velocity, acceleration, and angular acceleration information inside the stage solve. Lower-pair constraints enter at position, velocity, and acceleration levels through the stage residual. Endpoint velocity closure is audited through both the historical projection path and a square endpoint KKT residual.

At a schematic level, the accepted path solves

$$R_{\text{Gauss6}}(q_n, v_n, a_n, Z; h) = 0, \quad (1)$$

where Z denotes the FullVA stage variables. Lower-pair constraints appear as

$$\Phi(q) = 0, \quad \Phi_q(q)v = 0, \quad \Phi_q(q)a + \dot{\Phi}_q(q, v)v = 0. \quad (2)$$

The paper-TFE replacement target instead seeks a stage residual

$$R_{\text{TFE}}(q_n, v_n, a_n, Z; h) = [R_x \ R_y \ R_z \ R_{\text{NE}} \ R_{\Phi, v} \ R_{\Phi, a}]^T = 0, \quad (3)$$

with the same 132-row stage budget, but with rows derived from the paper temporal finite-element construction rather than from the Gauss stage formula. This replacement is counted as full only when the resulting trajectory solve, not just a one-step tangent or residual evaluation, passes the acceptance checks.

2.1 Accepted One-Step Integrator

The integrator claimed in Proposition 1 is the one-step map

$$x_{n+1} = \Psi_h^{\text{G6FVA}}(x_n), \quad x_n = (q_n, v_n, a_n), \quad (4)$$

defined by the accepted Gauss6/FullVA residual and its recorded endpoint closure audit. It is not the complete paper-TFE replacement residual. For each step, v047 uses the following evidence-backed construction:

1. assemble the cylindrical lower-pair constraints and FullVA stage unknowns Z for the current state x_n and step size h ;
2. solve the 132-row nonlinear stage system $R_{\text{Gauss6}}(q_n, v_n, a_n, Z; h) = 0$ to the Newton tolerance recorded in the CSV and JSON artifacts;
3. recover the raw endpoint by the Gauss collocation quadrature and the local Lie-group retraction used by the v047 trajectory code;
4. audit endpoint velocity closure through the square endpoint KKT path while keeping historical projection residual/correction measurements as diagnostics;
5. emit the per-step residual, constraint, order, runtime, and plotting artifacts consumed by the paper and pipeline validators.

The theorem endpoint object is the velocity-level/KKT closure functional. It is evaluated on the accepted Gauss6/FullVA path. Raw projection residuals are diagnostics, not theorem inputs or full-TFE replacements. This is the integrator whose smooth cylindrical-chain h -sweep records 7.161/7.066 position/velocity orders and whose four-example method gate is accepted.

2.2 Order Numbers

There are two order targets in the current evidence. The production method is the conditional order-six Gauss6/FullVA path, and the smooth benchmark measures position/velocity orders 7.161/7.066 on the tested range. The paper formula mapping audit separately encodes an $m = 3$ Gauss–Lobatto TFE operator with expected order five. An earlier source-free terminal-closed candidate has smooth orders 3.523/4.828. The latest endpoint-pose velocity predictor trajectory smoke closes terminal velocity even more tightly on the terminal row, but gives only 4.142/2.305 position/velocity order on the three- h smooth refinement. These are useful closure diagnostics, not accepted replacements, because they still degrade the trajectory order.

3 Proof Status

The proof ledger separates the load-bearing theorem from generated numerical evidence. The accepted Gauss6/FullVA path is not certified merely by a generic “Gauss collocation inherits order” slogan. The load-bearing argument fixes the implemented 132-row FullVA residual, inserts the classical three-stage Gauss local defect into that same stage object, uses the 96-row non-dynamic certificate and the 36-row Newton–Euler direct-substitution identity, and then passes the resulting $O(h^7)$ stage residual through the compact-tube Kantorovich, endpoint, inexact-Newton, and local-to-global transfer lemmas. Sparse AD and row/column coloring do not change this order because they only change how the same Newton residual and Jacobian are evaluated.

Assumption 1 (Smooth accepted-path order assumptions). *For the smooth cylindrical benchmark and reported h -sweep, assume:*

1. *the square FullVA residual is locally equivalent to a regular smooth constrained flow in a local Lie-group chart;*
2. *the three-stage Gauss collocation equations are solved to a Newton tolerance below the sixth-order truncation scale;*
3. *the local chart and retraction used for endpoint reconstruction are smooth diffeomorphisms on the tested trajectory tube;*
4. *the theorem endpoint output is the velocity-level/KKT closure functional on the same compact branch, while raw endpoint projection monitors remain diagnostics and do not supply the endpoint theorem input;*
5. *the sparse backend evaluates the same residual and Jacobian as the dense backend.*

Theorem 1 (Order of the accepted Gauss6/FullVA map). *Under Assumption 1, interpreted as the retained accepted-branch P1–P3/P6 compact-tube interfaces with the P4/P5 direct 96 + 36 residual bridge as the only consumed residual-value certificate, the accepted Gauss6/FullVA one-step map in Section 2.1 has local defect $O(h^7)$ and reduced-chart/reported position–velocity grid error $O(h^6)$ on fixed smooth time intervals, for reported grids whose transitions satisfy the predeclared branch, endpoint, proof-norm, and $\eta_h^{\text{tube}} \leq c_\eta h^7$ solver-envelope conditions. For driven or otherwise nonautonomous rows, the residual F_{A,h,t_n} , stage times $t_n + c_i h$, endpoint map, and proof norm are fixed on the same transition slice before constants or diagnostics are used. This is a conditional order-six statement for the accepted branch-selected map only. It retains P6 as a theorem-level solver-envelope condition, does not prove a P7 residual-to-error transfer, does not prove residual, reaction, multiplier, or constraint-output error estimates, does not use source-policy rows, and does not use or complete the separate primitive/Taylor T3 route. Equivalently, the theorem asserts only the local defect $O(h^7)$ and same-initial-state global error $O(h^6)$ under these retained interfaces and only for this displayed reduced/reporting grid scope.*

Proof. By Assumption 1, the proof is carried out on a fixed smooth compact branch in the local Lie-group chart. The classical three-stage Gauss local defect is inserted into the implemented FullVA stage object, not into a different reduced model. On that branch, the 96 non-dynamic rows provide the row-local $O(h^7)$ residual input and the 36 Newton–Euler rows vanish by direct substitution on the lifted Gauss stage. Thus the accepted 132-row bridge supplies the $O(h^7)$ stage residual defect used by the conditional theorem. Endpoint closure and the inexact Newton iterate add only $O(h^7)$ perturbations under the retained compact-tube hypotheses, and the local-to-global transfer then gives the $O(h^6)$ grid estimate for the same branch-selected reported grid. The proof order is the constrained Lie-group BDF order of obligations, not an estimate transfer: fix the discrete object, insert the smooth constrained branch, prove the local defect for that fixed object, and only then apply the stability recurrence and reporting map. No BDF local-error constant, hidden-constraint estimate, multiplier-history estimate, or multistep zero-stability theorem is imported into this one-step FullVA proof. The solver residual bound is a retained P6 envelope fixed before the run, not a conclusion drawn from fixed-tolerance logs, and the residual-to-error P7 interface remains open. Sparse and dense backends evaluate the same residual/Jacobian path, so backend choice is an implementation diagnostic, not an additional order argument. Residual-only mechanism rows, source-policy rows, and the separate primitive/Taylor T3 route are not theorem inputs and cannot be combined with the direct residual bridge to strengthen the theorem. $\square$

Lemma 1 (Order of the local paper-style TFE target). *The proof-level order target for the $m = 3$ Gauss–Lobatto paper-style TFE formula used in this manuscript is five.*

Proof. The source-paper family is compared through the temporal finite-element formula-order convention $p_{\text{TFE}}(m) = 2m - 1$. Substituting $m = 3$ gives $p_{\text{TFE}}(3) = 5$. This is distinct from the source paper’s reported DAE pendulum order reduction. $\square$

Theorem 2 (Accepted formal-order comparison theorem). *Assume Assumption 1. Also assume the current v047 artifact contract: the read-only validators accept the four-example coverage rows, the single- and double-pendulum dynamic rows provide the method-side order evidence, the four-link and slider-crank rows are treated only as closed-loop constraint/reaction checks, the smooth cylindrical h-sweep records the reported 7.161/7.066 position/velocity orders, the local paper-style $m = 3$ Lobatto-TFE formula target has expected order five, and the claim boundary records `full_tfe_stage_replacement=false`. Then the v047 evidence supports the following and only the following paper-facing statement: the accepted Gauss6/FullVA one-step map is a conditional sixth-order method relative to the local order-five paper-style TFE target and is covered by the four ASME-style examples under the stated evidence boundary. The theorem does not assert complete source-paper TFE residual reproduction.*

Proof. Order. Theorem 1 gives the conditional order-six result for the accepted Gauss6/FullVA one-step map through the direct 132-row residual bridge under the smooth compact-branch and solver-scale interfaces. The generated convergence artifacts give the independent numerical check used by the paper: the smooth cylindrical h-sweep records position and velocity orders 7.161/7.066.

Four-example evidence. The same artifact contract accepts the single pendulum, double pendulum, four link, and slider crank method rows on the Gauss6/FullVA path. These rows are accepted as method evidence rows, not as full-TFE replacement rows.

Comparison. The comparator in this paper is the locally encoded paper-style $m = 3$ Lobatto-TFE formula target. Lemma 1 fixes its expected order as five. Combining this order-five comparator with the conditional order-six accepted Gauss6/FullVA map proves the formal-order comparison statement inside the stated artifact scope.

Boundary. A complete source-paper residual reproduction would require accepted paper-derived stage rows inside the 132-row nonlinear solve, without endpoint source data, terminal-row replacement, projection, or a target-direction oracle. The current artifact contract records `full_tfe_stage_replacement=false`; therefore that stronger statement is excluded rather than silently assumed. The theorem proves only the accepted formal-order comparison claim. $\square$

Corollary 1 (Artifact-backed formal-order comparison acceptance). *Under Assumption 1 and the current read-only v047 artifact checks, Proposition 1 is accepted within the stated artifact scope.*

Proof. Theorem 1 gives the conditional order-six direct residual-bridge theorem for the accepted Gauss6/FullVA one-step map, and Theorem 2 packages the order, four-example, comparator, and boundary evidence into the formal paper-facing claim. The generated convergence artifacts record smooth position/velocity orders 7.161/7.066, and the minimal four-example validator accepts single pendulum, double pendulum, four link, and slider crank on the accepted method path. The comparator definition keeps the local $m = 3$ Lobatto-TFE expected-order-five target separate from the complete source-paper residual reproduction gate. The full-TFE gap and repair-spec validators keep `full_tfe_stage_replacement=false` explicit, so the corollary accepts only the formal-order comparison and not the stronger reproduction claim. $\square$

This is deliberately not a proof of the independent paper-TFE replacement. The current paper-TFE rows are either formula maps, local tangent probes, or diagnostic substitutions unless they replace the stage residual inside the nonlinear h-sweep and pass the acceptance test in Table 15. The velocity-level/KKT closure functional is the theorem endpoint object on the accepted Gauss path; raw endpoint projection measurements are diagnostics and do not discharge the endpoint raw-defect hypothesis or define an accepted full-TFE method.

4 Validation Protocol

Every reported method claim is tied to generated artifacts in the v047 results directory. The core h-sweeps use three step sizes $h \in \{0.04, 0.02, 0.01\}$ against an $h = 0.005$ reference unless otherwise noted. The four ASME-style examples are:

- single pendulum;
- double pendulum;
- four link;
- slider crank.

The validation records CSV rows, summary JSON entries, plots, and a generated report. Accepted method rows require small residuals, clean constraints, and consistent observed order or a documented reference policy. Diagnostic rows are kept separate from accepted rows when they use projection, endpoint-source data, local tangent probes, terminal-row replacement, or incomplete TFE stage substitution.

4.1 Proposition Evidence Traceability

Table 9 records the evidence chain for Proposition 1. It is deliberately read-only: every entry is checked from existing CSV, JSON, PDF, and ledger artifacts rather than by rerunning the full v047 generator.

5 Accepted Gauss6/FullVA Results

Table 10 gives the headline convergence behavior. The smooth branch is above sixth order on the tested range. The sharp branch is not treated as a silent failure; it is tracked by a Brown–McPhee smoothness sweep and by deeper fixed refinements.

The ultra-refinement audit recovers high observed order, but only at increased work. Relative to the $h = 0.01$ sharp baseline, the high-order window around $h = 0.00125$ costs about $7.25 \times$ runtime and $8 \times$ steps/Newton iterations while reducing velocity error by approximately 1.06×10^6 . The remaining sharp-friction caveat is therefore a practical coarse-regime cost issue rather than a complete asymptotic failure.

6 Four-Example Mechanism Gate

The four-example gate is accepted as coverage, not as four independent dynamic-order proofs. This does not mean all research caveats are closed: sparse speed, the independent full-TFE stage replacement, and practical sharp-friction coarse-step behavior remain open items. The machine-readable

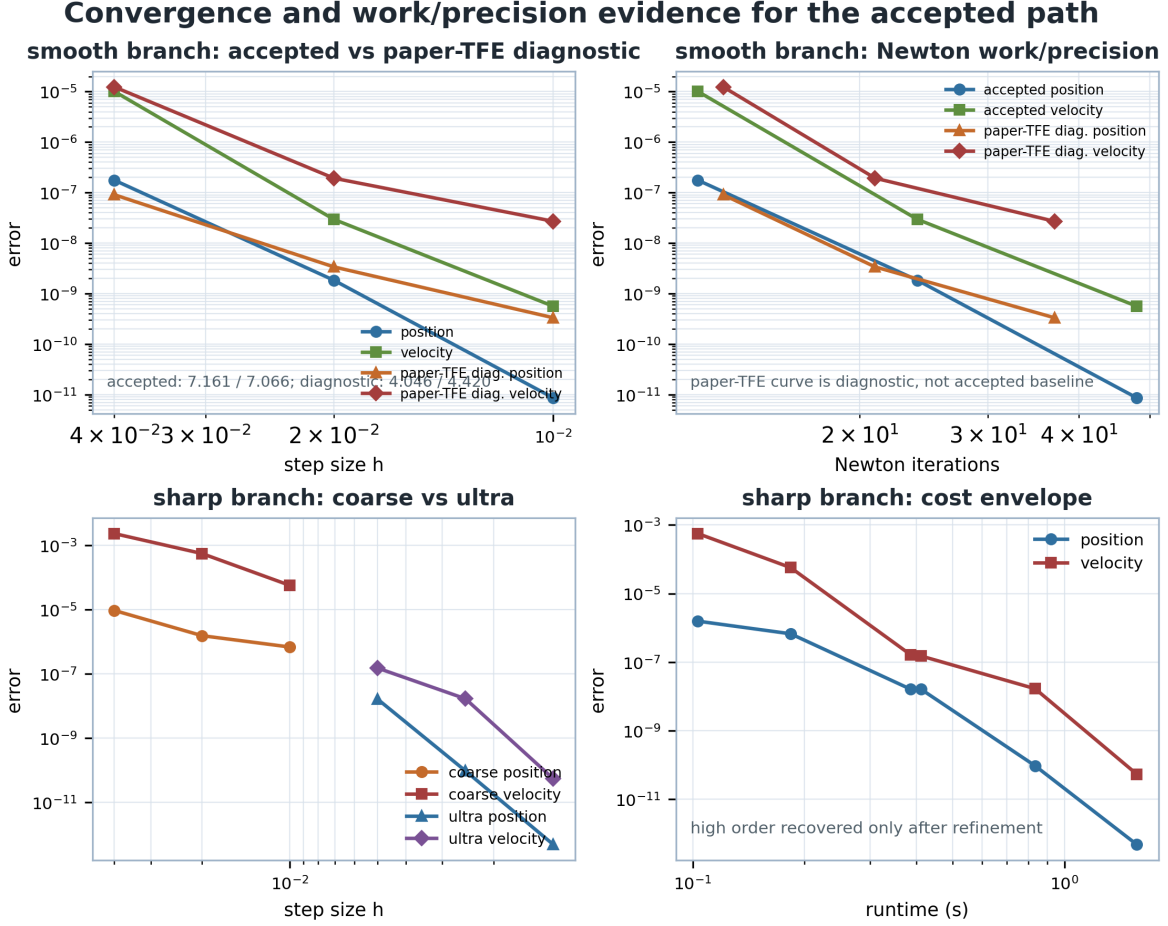


Figure 1: Cylindrical-chain convergence for the accepted Gauss6/FullVA path. The smooth case is high order; the sharp-friction case is order-reduced on the coarse h -sweep.

summary status for this claim is `four_asme_method_rows_accepted_projection_sharp_sparse_caveats`.

7 Endpoint Closure and TFE Replacement Audit

The endpoint KKT diagnostic closes endpoint velocity to roundoff while preserving the accepted Gauss6/FullVA stage equations. It is important evidence, but it is not the same as a complete paper-TFE stage replacement. The full TFE replacement target is stricter: all 132 stage rows must be replaced by independently derived paper-TFE rows, and the resulting nonlinear h -sweep must pass residual, rank, endpoint, and observed-order checks without relying on endpoint projection, endpoint source data, terminal-row substitution, or local tangent-only arguments.

7.1 Latest Source-Free Final-Stage Closure

The latest lower-pair audit inserts 16 centered acceleration source-consistency rows and 8 final paper-stage velocity rows. It removes the external endpoint source, removes endpoint-boundary source data from the trajectory, avoids terminal-row replacement, and does not use velocity pro-

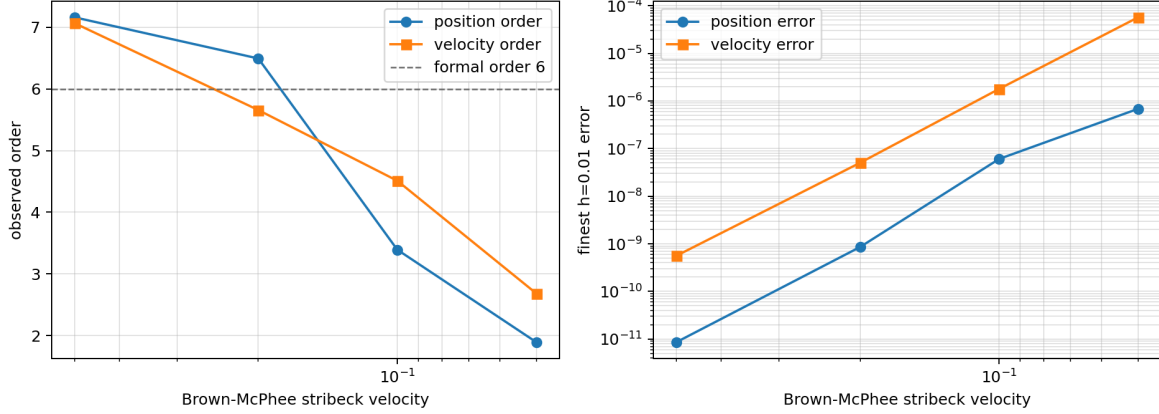


Figure 2: Brown-McPhee smoothness sweep. Reducing the stribek velocity moves the problem from a smooth high-order branch to a coarse sharp-friction order-reduced branch.

jection. The nonlinear h -sweep solves six smooth/sharp trajectory rows with maximum residual 9.49×10^{-12} and raw terminal endpoint velocity 4.01×10^{-16} .

This is a real improvement: the terminal-velocity gate is closed by source-free stage rows. It is still not an accepted full-TFE replacement because the smooth position order is 3.523, below the acceptance floor used by the pipeline. The current blocker has shifted from terminal closure to preserving paper-TFE order after the closure rows are inserted into the nonlinear trajectory solve.

7.2 Latest Endpoint-Pose Trajectory Smoke

The latest bounded JSON-only follow-up promotes the local endpoint-pose row into the nonlinear 132-row residual without running the full artifact generator. The target `lower_pair_endpoint_pose_velocity_predictor_h_sweep_smoke` keeps rank 132 and converges all requested smooth short-trajectory steps. With the terminal-equivalent `stage02_convex_pose_velocity_0p00_z` row, the three- h refinement over $h = [0.04, 0.02, 0.01]$ against reference $h = 0.005$ closes raw terminal endpoint velocity to 8.124×10^{-17} , but the position/velocity orders are only 4.142/2.305.

This resolves an ambiguity in the local row-space evidence. The endpoint row is not merely untested at trajectory level: it closes terminal velocity inside Newton, but loses the paper-TFE order. Moving slightly away from the endpoint does not fix the problem. The nearby nonterminal `stage02_convex_pose_velocity_0p01_z` row converges, but leaves terminal endpoint velocity at 6.255×10^{-6} and gives only 2.345/1.878 orders. The remaining repair target is therefore a non-terminal endpoint-pose predictor that keeps terminal velocity closed on trajectory while recovering the paper-TFE order target.

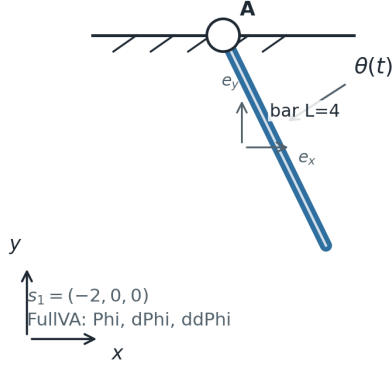
The best non-projection follow-up in this family is the terminal-limit extrapolated row `stage02_convex_pose_velocity_extrapolate_0p01_0p02_z`. The default two- h smoke has `projection_used=false`, reduces terminal velocity to 1.254×10^{-7} , and records 6.412/3.775 orders. However, the three- h smooth refinement gives only 4.054/2.364 position/velocity order and still has maximum terminal velocity 1.254×10^{-7} , with the finest- h row at 1.244×10^{-8} . It is therefore a useful trajectory near-miss, not an accepted replacement.

Moving the extrapolation points closer to the terminal endpoint improves only the terminal velocity. The default two- h sweep over `0p005/0p01`, `0p002/0p005`, and `0p001/0p002` has `projection_used=false`, terminal velocities 3.135×10^{-8} , 6.270×10^{-9} , and 1.254×10^{-9} , and

ASME-style lower-pair validation mechanisms

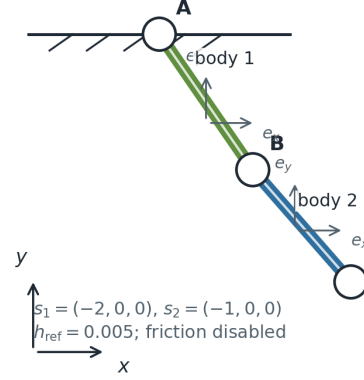
single pendulum

CD(3) + DP1(3); driven theta(t)



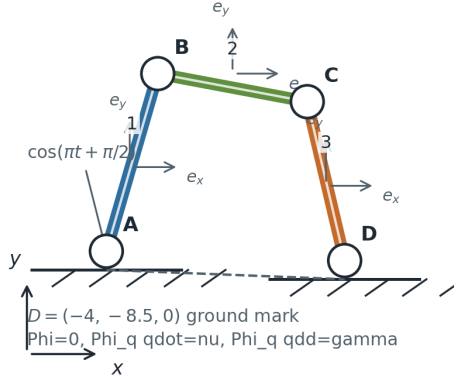
double pendulum

CD(6) + DP1(4); nested FullVA reference



four-link loop

CD(12) + DP1(6); driven closed loop



slider-crank

CD(6) + DP1(7) + DP2(4) + D(1)

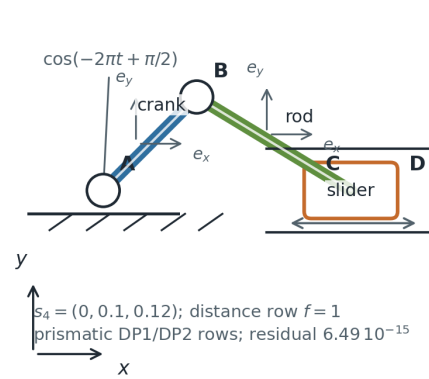


Figure 3: Lower-pair graph bridge over the four ASME-style examples.

orders 6.621/4.648, 6.599/4.566, and 6.591/4.521. A three- h refinement of `stage02_convex_pose_velocity_extrapolate_0p001_0p002_z` gives terminal velocities 1.254×10^{-9} overall and 1.244×10^{-10} at the finest step, but only 4.143/2.307 order. The ultra-near `stage02_convex_pose_velocity_extrapolate_0p00001_0p00002_z` row finally satisfies the terminal-velocity tolerance without projection, with max terminal velocity 1.254×10^{-13} and finest- h terminal velocity 1.254×10^{-14} , but its three- h order is still 4.142/2.305 and `smooth_order_ok=false`. This is terminal-equivalent limiting evidence, not a full-TFE replacement.

A three-point quadratic extrapolation check reaches the same conclusion without evaluating the terminal row. The `stage02_convex_pose_velocity_quadextrapolate_*` laws estimate the weight-zero closure from three strictly nonterminal rows and keep `projection_used=false`. The default two- h sweep over `0p01/0p02/0p05`, `0p005/0p01/0p02`, and `0p002/0p005/0p01` records terminal velocities 2.029×10^{-11} , 2.029×10^{-12} , and 2.029×10^{-13} , but all three velocity orders remain about 4.508. The best row, `stage02_convex_pose_velocity_quadextrapolate_0p002_0p005_0p01_z`, closes the terminal-velocity tolerance on the three- h refinement, with max terminal velocity 2.029×10^{-13} and finest- h terminal velocity 8.254×10^{-15} , but its order is still 4.142/2.305. Thus quadratic nonterminal extrapolation is also terminal-equivalent limiting evidence rather than

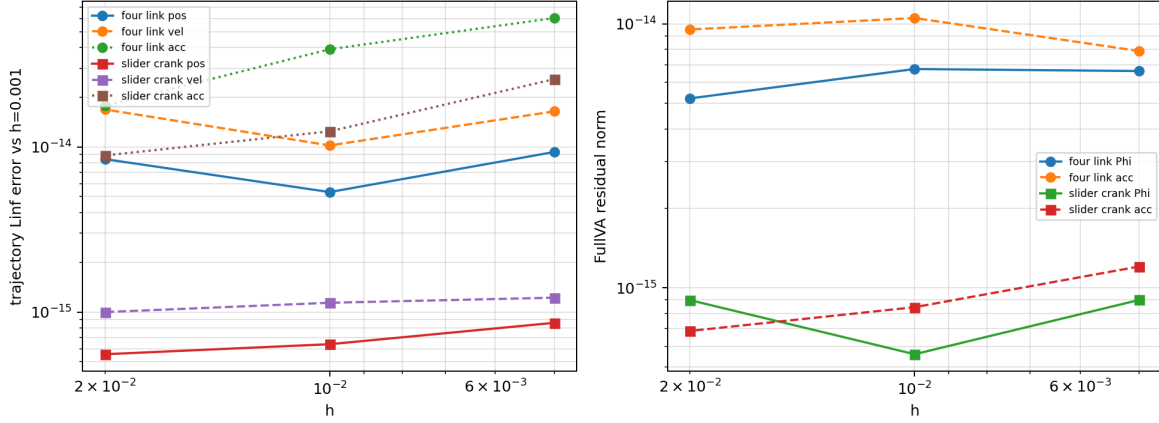


Figure 4: Closed-loop kinematic FullVA solves for the four-link and slider-crank examples.

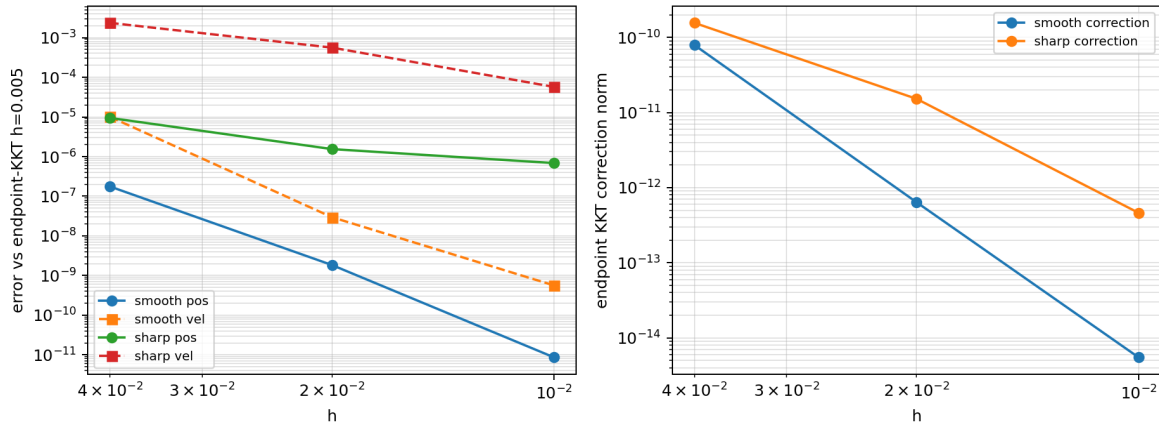


Figure 5: Endpoint KKT residual closure. The endpoint velocity constraint is enforced inside a square residual in this diagnostic path.

a full-TFE replacement.

A projection-like terminal-tangent follow-up rules out the simplest correction of that nonterminal row. The full `stage02_convex_pose_velocity_0p01_terminalproj_z` correction fails in the reference run with residual divergence 6.739×10^8 . Damped variants converge but move in the wrong direction: `stage02_convex_pose_velocity_0p01_terminalproj_p0p1_z` leaves terminal velocity at 7.028×10^{-6} with orders 2.301/1.788, and `stage02_convex_pose_velocity_0p01_terminalproj_p0p5_z` leaves terminal velocity at 1.321×10^{-5} with orders 2.158/1.433. These rows set `projection_used=true`, so they are diagnostic exclusions rather than candidate acceptance evidence.

7.3 Why the Next Audit Is an Eight-Row Velocity Compression

The source-free elimination audit shows that the 24 lower-pair stage rows have rank 16 after removing the external source; the missing rank is exactly eight rows. Mean-velocity and scalar/component blend closures preserve useful order but leave terminal endpoint velocity at roughly 10^{-5} . The final-stage velocity closure drives terminal velocity to roundoff, but its smooth position order drops to

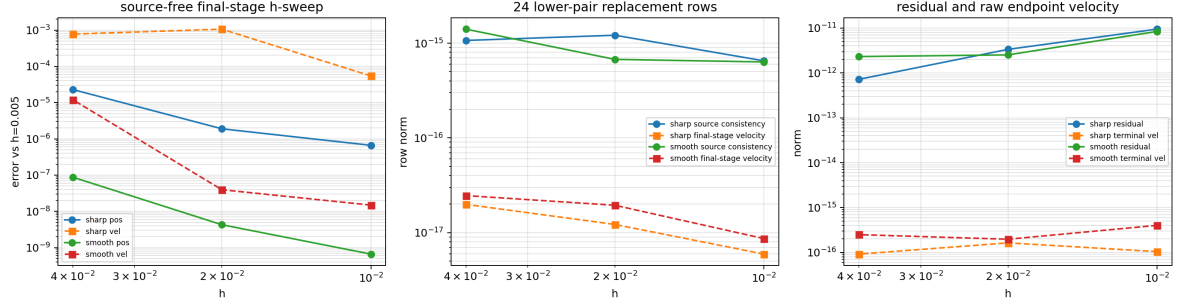


Figure 6: Source-free final-stage lower-pair velocity closure. This audit closes raw terminal endpoint velocity but remains smooth-position-order limited.

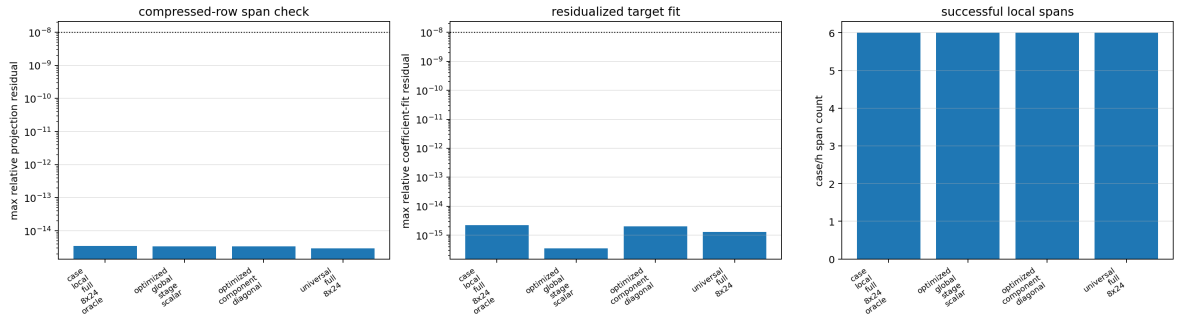


Figure 7: Velocity-compression audit. The all-stage lower-pair velocity rows span the terminal bridge in the local tangent model, but nonlinear trajectory acceptance of a universal eight-row compression remains pending.

3.523. The problem is therefore not the existence of any closure row, but the choice of the eight source-free rows that close terminal velocity while preserving the paper-stage order.

The current velocity-compression audit is intentionally local. It tests 24 candidate/case/step tangent rows and finds that fixed source-free compression can span the terminal bridge directions, with the best fixed candidate recording projection residuals around 10^{-15} . The strengthened audit now shows that those fixed compressions collapse to the final-stage selector: the universal full 8×24 selector-relative distance is 1.53×10^{-15} and the largest non-final-stage energy fraction is 1.71×10^{-15} .

The follow-up non-degenerate probe makes this failure mode explicit. It sweeps 192 candidate/case/step/eta rows by fixing the final-stage coefficient at one and injecting stage-0 and/or stage-1 lower-pair velocity rows. There are 84 rows with meaningful non-final-stage energy, but none of them spans the local terminal-bridge target; the best meaningful projection residual is 1.59×10^{-3} , far above the 10^{-8} local-span threshold. Only the eta-zero rows span, and their non-final-stage energy is exactly zero. Thus the local compression is not yet a new independent closure formula.

The state-dependent follow-up tests the natural next option: choose diagonal stage directions from the local tangent state rather than using fixed scalar stage weights. It sweeps 72 candidate/case/step/eta rows over a 65-direction finite grid. There are 36 rows with meaningful non-final-stage energy, but none spans the local terminal-bridge target; the best meaningful projection residual is 9.90×10^{-3} . This rules out diagonal state-dependent direction weights as the missing

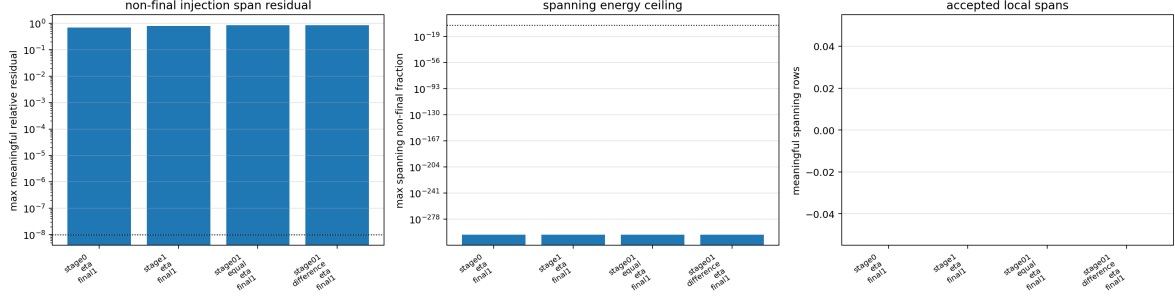


Figure 8: Non-degenerate velocity-compression probe. Fixed stage-0/stage-1 injections with meaningful non-final-stage energy lose the local terminal-bridge span.

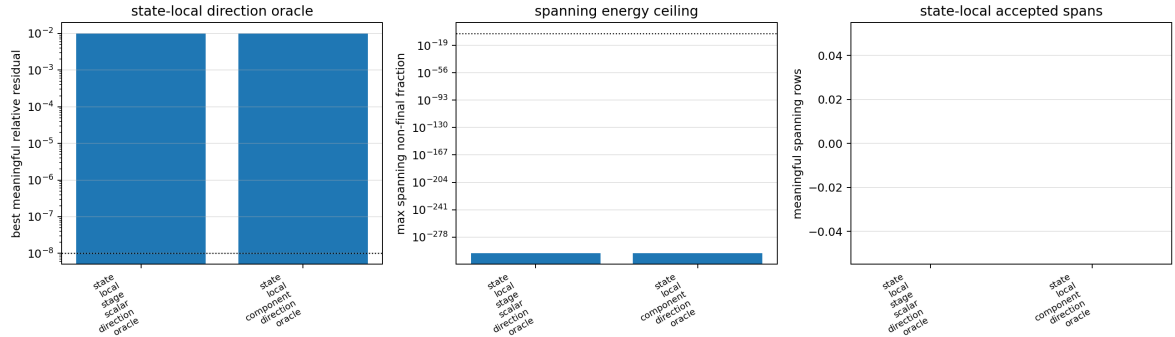


Figure 9: State-dependent velocity-compression probe. Diagonal state-local direction oracles with meaningful non-final-stage energy still fail to span the local terminal-bridge target.

independent closure.

The full component-mixing follow-up removes the final-stage velocity rows from the active fit and allows full 8×8 or 8×16 mixing over stage 0, stage 1, or both non-final stages. It tests 36 case/step/candidate rows using both case-local and universal fits. None spans the local terminal-bridge target; the best tangent residual is 9.81×10^{-1} . Some rows are value-balanced to roundoff, with minimum value residual 2.19×10^{-16} , but value balance alone is not a tangent closure. The result motivates the value-level and derivative-aware probes below, rather than serving as an accepted full-TFE replacement.

The value-level follow-up reverses the diagnostic priority: it fits row values first, then checks whether the induced Jacobian rows span the terminal bridge. It tests 72 case/step/candidate rows. All 72 rows balance values; 18 also span the local tangent. However, the spanning value-balanced rows are final-stage-selector-like: the maximum spanning non-final-stage energy fraction is 2.12×10^{-15} , and the count of meaningful non-final value-balanced tangent spans is zero. This rules out a purely value-fitted non-final closure as the missing independent replacement. The next target is a meaningful non-final, derivative-aware value-level compression.

The derivative-aware follow-up keeps the same value-level non-final row values and asks whether a local coefficient-gradient correction can close the tangent gap. It tests 36 case/step/candidate rows over scalar, component-wise, and full non-final active velocity families. All 36 rows balance values and span the local terminal-bridge tangent with meaningful non-final energy after the oracle correction. The best corrected projection residual is 9.02×10^{-16} , while the best uncorrected value-level base residual is still 9.82×10^{-1} and the maximum required coefficient-gradient Frobenius

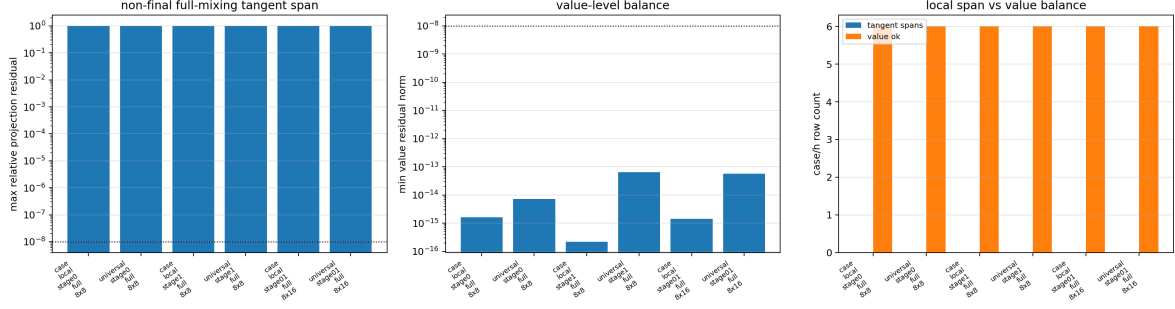


Figure 10: Non-final component-mixing probe. Removing the final-stage rows and allowing full stage-0/stage-1 component mixing still does not span the terminal bridge.

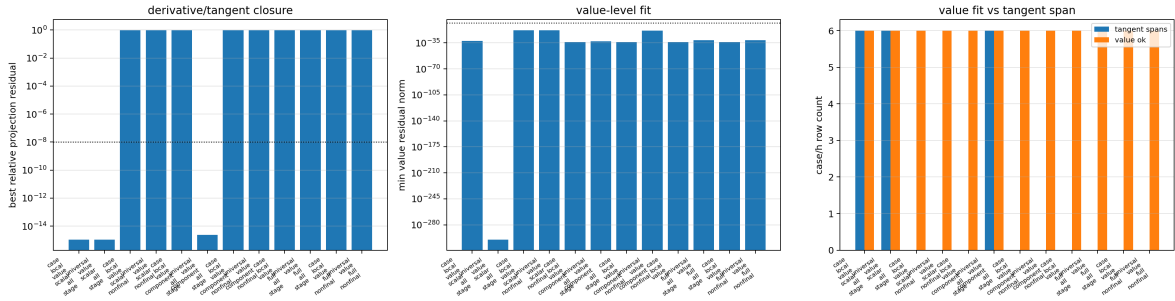


Figure 11: Value-level source-free compression probe. Value fits can balance rows, but meaningful non-final value-level fits do not span the terminal bridge.

norm is 1.95×10^{17} . This is a useful upper bound: it says the next repair can focus on deriving the missing coefficient-gradient law rather than searching for another value-only row family. It is not yet a full TFE replacement, because the correction has not been converted into a bounded analytic formula and inserted into a nonlinear trajectory h-sweep.

The bounded-gradient follow-up quantifies how hard the oracle is to realize as a formula. It tests 288 cap/candidate/case/step rows. All 288 rows preserve the value balance and meaningful non-final row energy, but only 92 rows span at the tested caps. A practical cap of 10^{12} spans only 12 rows; all 36 local derivative-aware spans appear only at cap 10^{18} . Thus the local tangent gap is closed only by a very large derivative budget. The next mathematical target is still a bounded analytic coefficient-gradient law followed by a nonlinear h-sweep, not another unconstrained local fit.

The bounded-formula follow-up replaces the hard cap with explicit saturation laws: global tanh, rowwise tanh, and rowwise rational-quadratic scaling. It tests 648 law/cap/candidate/case/step rows. All 648 rows preserve value balance and meaningful non-final row energy, and 366 rows span at the tested caps. For each tested law, complete coverage of the 36 local derivative-aware spans appears only at cap 10^{22} . The best single-row projection residual is 8.02×10^{-16} , attained by the rowwise tanh law at cap 10^{14} . This is still not a full TFE replacement: the saturation formula bounds the magnitude, but the correction direction remains the local target-direction oracle and no nonlinear trajectory h-sweep has accepted it.

The target-direction-free follow-up is the narrowest diagnostic needed after the saturation-law result. It tests 2592 law/cap/candidate/case/step rows over 12 direction laws: the positive and neg-

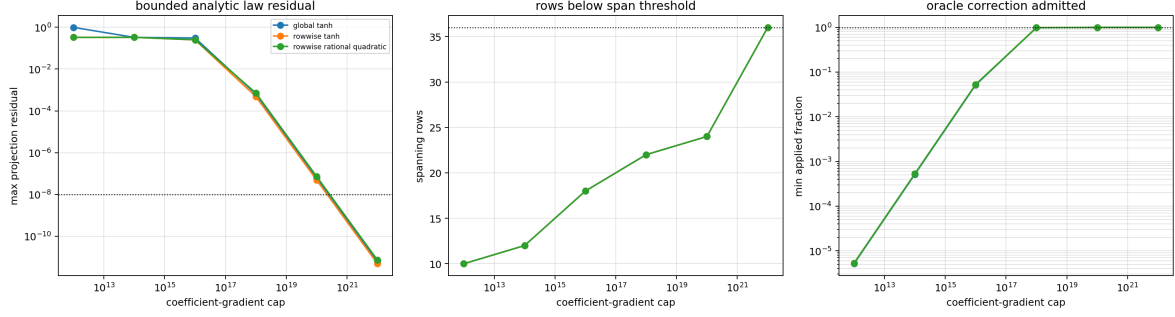


Figure 14: Bounded analytic saturation-law probe. Tanh and rational saturation laws preserve the value balance, but all local spans require cap 10^{22} and the correction direction is still a target-direction oracle.

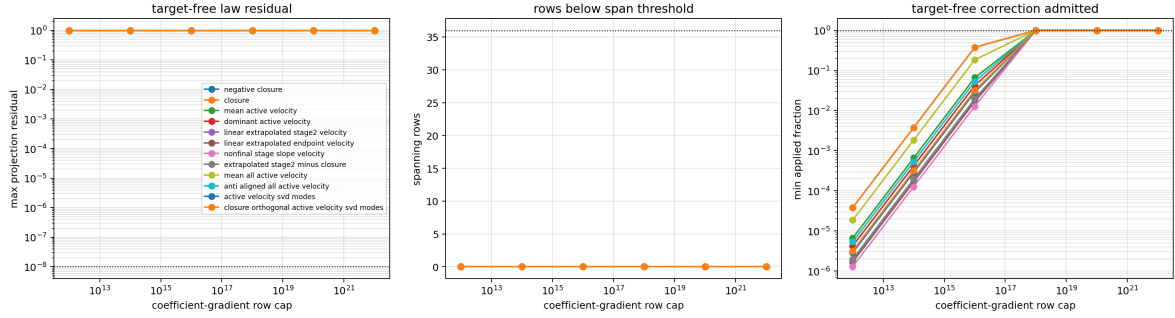


Figure 15: Target-direction-free bounded formula probe. The correction direction is computed from the current closure and active non-final velocity rows, not from the target Jacobian.

test from linear feature capacity to nonlinear second-differential source features.

The nonlinear-capacity audit tests 216 rows across six finite-difference source-feature families. It records zero spanning rows: the best projection residual is 6.95×10^{-1} and the best correction residual is 7.01×10^{-1} . The target Jacobian is used only to define and project the local correction being tested, not to construct a target-direction oracle formula. No final-stage velocity selector is accepted, no nonlinear h-sweep is accepted, and the tested second-differential non-final source features are ruled out. This motivates the higher-order and nonlocal source-feature test below.

The higher-order/nonlocal capacity audit tests another 216 rows across six feature families: third differentials along active non-final velocity rows, active-plus-third-differential rows, second-plus-third source differentials, same-case cross-step source-row deltas, active-plus-nonlocal deltas, and a combined higher-order/nonlocal source feature library. It again records zero spanning rows. The best projection residual is 6.95×10^{-1} , and the best correction residual is 7.01×10^{-1} . This closes the narrow next step from the previous audit: the missing coefficient-gradient law is not present in these tested third-differential or simple cross-step nonlocal source features. The next repair target is therefore a different history-dependent non-final source law or a revised weak-row formula, followed by the nonlinear h-sweep.

The history-transport capacity audit tests another 216 rows across six feature families: transported source-row deltas, active-plus-source transport deltas, transported non-final velocity-row deltas, active-plus-velocity transport deltas, combined source/velocity transport deltas, and the same combined transport library with active non-final velocity rows included. It again records zero

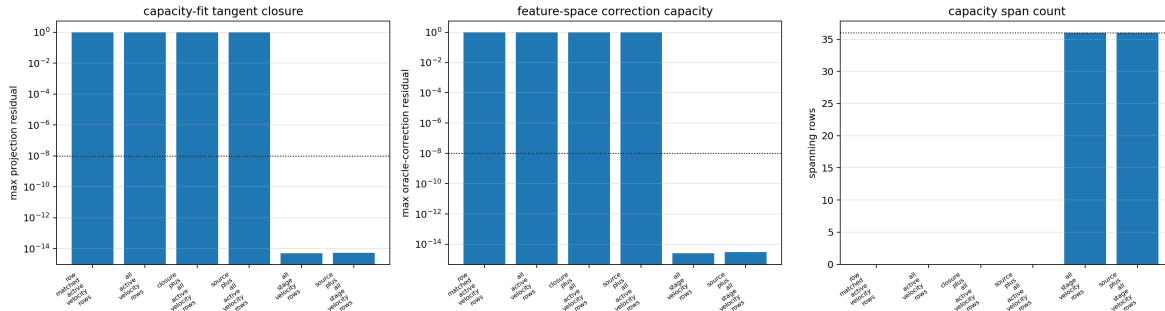


Figure 16: Direction-capacity audit. All-stage velocity feature spaces can reproduce the local derivative-aware correction, but non-final active/source feature spaces cannot.

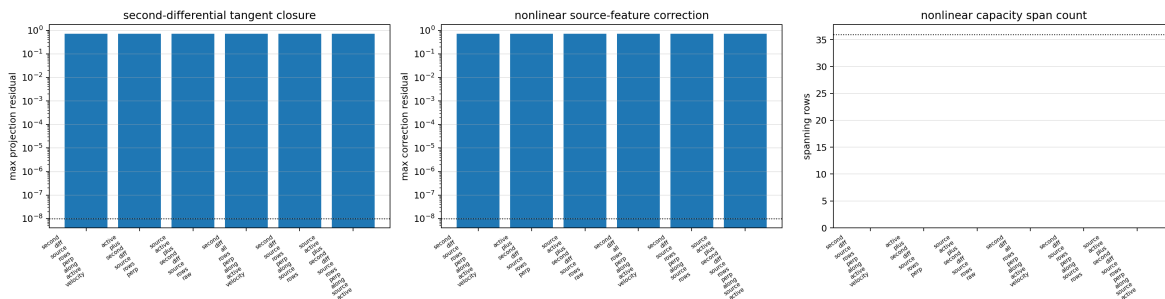


Figure 17: Nonlinear second-differential capacity audit. Finite-difference source features increase the local non-final feature library, but none spans the local lower-pair correction.

spanning rows. The best projection residual is 7.13×10^{-1} , and the best correction residual is 7.12×10^{-1} . Thus a one-step trajectory-history feature space is not enough.

The two-step history-transport capacity audit advances the terminal output twice and stacks the transported source-row and non-final velocity-row deltas from both future states. It again tests 216 rows across six feature families and records zero spans. The best projection residual is 6.95×10^{-1} , and the best correction residual is 7.01×10^{-1} . The remaining target is therefore no longer simply “try history”: the tested one-step and two-step transported feature spaces are insufficient.

The recurrent history-capacity audit then tests 216 rows over two-step source/velocity curvature and pairwise bilinear source/velocity history features. It records zero spans; the best projection and correction residuals are 6.952×10^{-1} and 7.012×10^{-1} . The next repair therefore needs a revised weak-row formula or richer nonlinear recurrent history source law, followed by the nonlinear h-sweep.

The weak-row structure-capacity audit then tests 216 rows over six algebraic source-active feature families. These features are target-free in construction and use the target Jacobian only for capacity scoring. They again record zero spans; the best projection and correction residuals are 6.952×10^{-1} and 7.012×10^{-1} . This is a narrower negative result than the history tests: the missing row is not recovered by the tested cross-Gram, Hadamard, cross-Hadamard, or shifted-commutator structure rows. The next repair therefore needs a revised analytical weak-row formula or a richer nonlinear recurrent history source law.

The row-space compression audit removes the target Jacobian and target direction from the

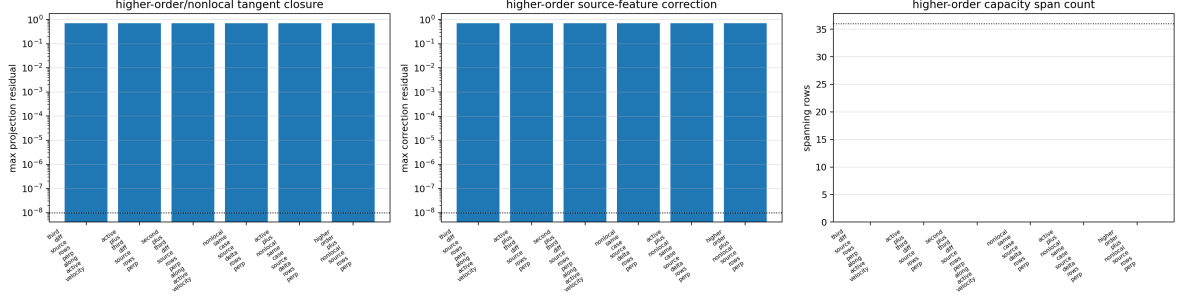


Figure 18: Higher-order/nonlocal capacity audit. Third finite-difference source features and same-case cross-step source deltas still do not span the local lower-pair correction.

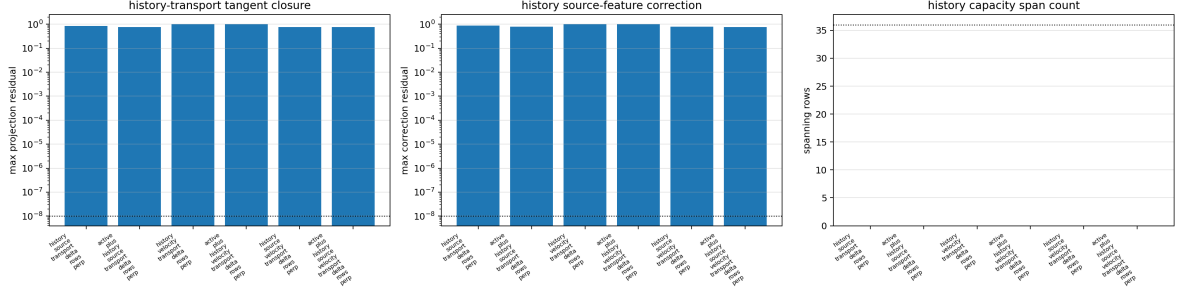


Figure 19: History-transport capacity audit. One-step transported source-row and non-final velocity-row deltas still do not span the local lower-pair correction.

formula construction. It builds frozen case-local compression matrices from the velocity row space only, then scores whether the resulting eight rows span the terminal lower-pair bridge. Across 36 rows covering six laws, two cases, and three step sizes, it records zero spans and 24 value-balanced rows. The best projection residual is 7.378×10^{-1} for `all_stage_svd_top8`; the best value residual is 3.68×10^{-12} for the same law, but that value balance does not close the tangent gap. The best non-final residual is 9.815×10^{-1} . Because the coefficients are frozen state-dependent quantities and coefficient derivatives are not included, this is not a nonlinear full-TFE residual. It rules out this target-free row-space compression family and leaves the same repair target: a revised analytical weak-row formula or richer nonlinear recurrent history source law.

That targeted promotion has now been run as an 18-row diagnostic over two cases, three step sizes, and beta values $\{0, 0.5, 1\}$. The beta-zero and beta-half smooth branches preserve position order above five, but leave raw terminal endpoint velocity around 4×10^{-6} . The beta-one branch closes terminal velocity to 4.01×10^{-16} , but has smooth position order 3.523. Thus the intersection flag is false: no tested beta simultaneously satisfies the terminal-velocity and order gates. The evidence narrows the next repair to a genuinely new eight-row source-free closure formula rather than another scalar blend on the same endpoints.

A newer recurrent/terminal gamma-blend smoke promotes the same tradeoff to a bounded trajectory h-sweep without invoking the full generator. The latest default run used smooth and sharp cases, $\gamma \in \{0.5, 1.0\}$, $h \in \{0.04, 0.02\}$, and an $h = 0.01$ reference. It converged all 24 requested steps with rank 132 and maximum residual 4.77×10^{-12} . The terminal-closing branch is still $\gamma = 1.0$, with raw terminal endpoint velocities 3.05×10^{-16} and 2.94×10^{-16} in the smooth and sharp cases, but its smooth minimum order is only 4.397 and the sharp velocity order remains

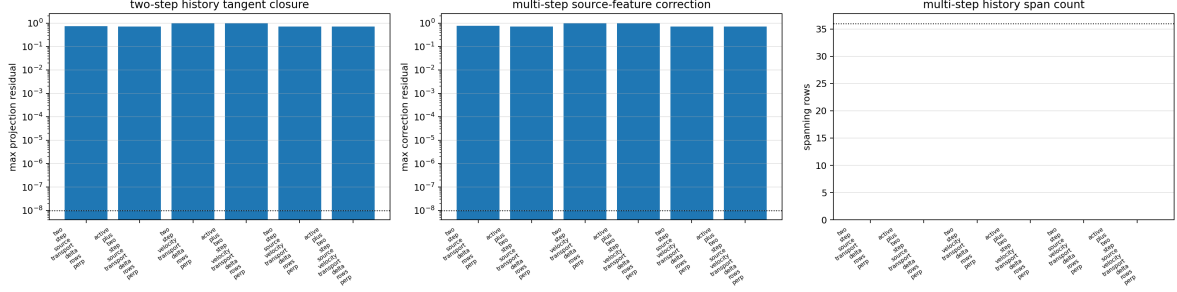


Figure 20: Two-step history-transport capacity audit. Transported source-row and non-final velocity-row deltas from two future states still do not span the local lower-pair correction.

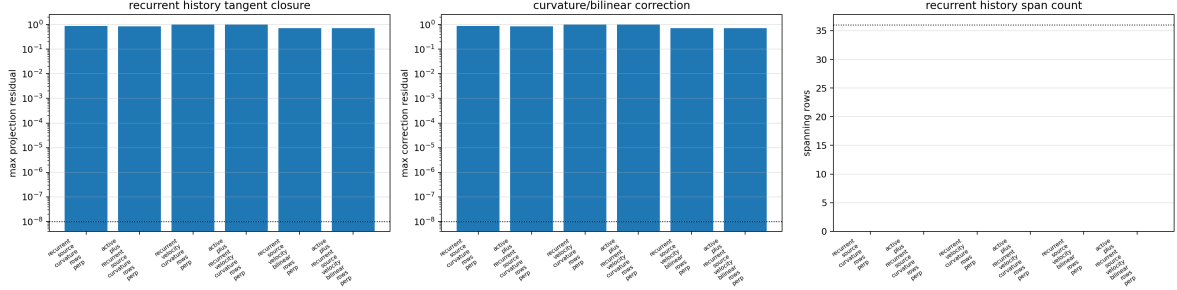


Figure 21: Recurrent history-capacity audit. Two-step curvature and bilinear source/velocity history features still do not span the local lower-pair correction.

-0.428 . The best smooth-order branch, $\gamma = 0.5$, reaches minimum order 4.413 but leaves terminal velocity up to 8.11×10^{-6} on the sharp case. This targeted result is consistent with the larger order/closure audit: the properties remain split, so the full-TFE gate stays open.

A denser bounded scalar-gamma screen over $\gamma \in \{0, 0.25, 0.5, 0.75, 1\}$ converged all 60 requested short-trajectory steps with the same rank-132 residual structure. It did not change the conclusion. The best smooth-order branch moved only to $\gamma = 0.75$, with minimum order 4.414, while terminal closure remained only on the $\gamma = 1.0$ branch and the maximum terminal velocity over the grid stayed at 8.23×10^{-6} . The scalar interpolation family is therefore exhausted as a plausible repair; the next candidate must change the eight lower-pair closure rows themselves.

A near-terminal scalar-policy screen over $\gamma \in \{0.9, 0.99, 0.999, 1\}$ also fails. It converges all 48 requested short-trajectory steps, and its best smooth-order branch is $\gamma = 0.99$ with minimum order 4.415. However, that branch leaves sharp terminal velocity at 3.08×10^{-6} , while even $\gamma = 0.999$ leaves 4.68×10^{-7} . Only $\gamma = 1.0$ closes terminal velocity to roundoff, and that is exactly the order-limited final-stage endpoint. This closes the near-terminal scalar-policy loophole.

A component-wise one-step gamma sensitivity check then releases one of the eight terminal directions at a time to $\gamma = 0.999$. The run covers smooth and sharp cases with 10 gamma vectors, for 20 one-step rows, and completes in 31.6 seconds. It preserves rank 132 with maximum residual 1.77×10^{-12} . Five nontrivial rows close terminal velocity at the 10^{-12} tolerance; the best is `component_2_release_0p999` at 3.46×10^{-17} , while `component_7_release_0p999` is the worst component release at 1.23×10^{-11} . This is useful sensitivity evidence, but it is still a one-step screen rather than a trajectory h-sweep, so the independent full-TFE replacement remains open.

The same component idea was then promoted to a bounded trajectory h-sweep over four vec-

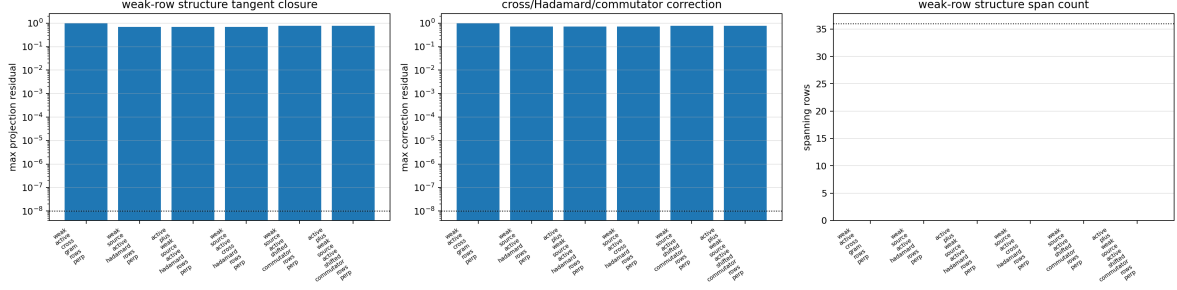


Figure 22: Weak-row structure-capacity audit. Source/active cross-Gram, Hadamard, cross-Hadamard, and shifted-commutator feature rows still do not span the local lower-pair correction.

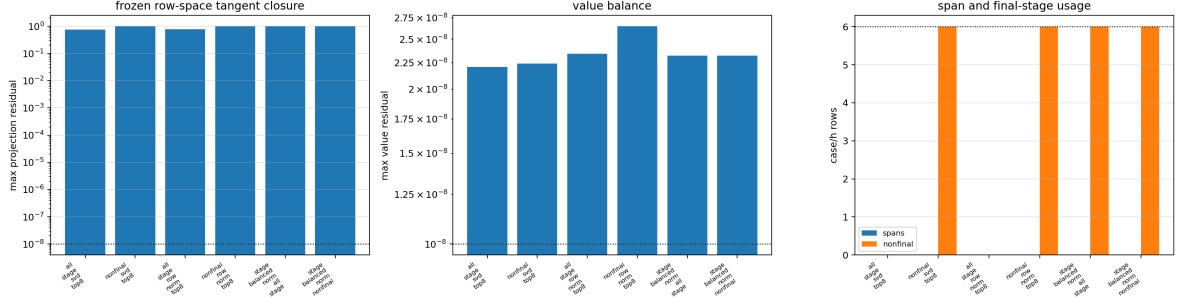


Figure 23: Row-space compression audit. Target-free SVD, row-norm, and stage-balanced frozen eight-row compressions of the all-stage/non-final velocity row space do not provide a value-balanced tangent closure.

tor labels: `all_ones`, `all_0p999`, `component_2_release_0p999`, and `component_7_release_0p999`. The run covers smooth and sharp cases, $h \in \{0.04, 0.02\}$, and an $h = 0.01$ reference, converging all 48 short trajectory steps in 98.4 seconds with rank 132 and maximum residual 4.77×10^{-12} . The best terminal vector is `component_2_release_0p999`, with terminal velocity 2.70×10^{-16} , but its smooth minimum order remains about 4.397. The best smooth-order vector is `all_0p999`, with smooth minimum order 4.401, but it leaves terminal velocity 4.68×10^{-7} . Thus the trajectory-level component screen still has no order/terminal intersection.

A final lightweight nonlinear-source smoke sets the component-wise blend from the previous step's history source, rather than from a prescribed scalar or fixed vector. The default smooth-only `history_abs_inf` run uses releases $\{0.001, 0.01\}$, the same $h \in \{0.04, 0.02\}$ sweep, and an $h = 0.01$ reference. It converges all 12 short-trajectory steps in 38.9 seconds with rank 132 and maximum residual 2.59×10^{-12} . The best branch has terminal velocity 1.45×10^{-7} and smooth minimum order 4.402. A signed-policy follow-up with `history_signed_positive` and `history_signed_negative` converges in 36.9 seconds; its best terminal velocity is still 1.45×10^{-7} and its best smooth minimum order is 4.415. This excludes the simple explicit recurrent history-gamma law as the missing full-TFE replacement. The next target is therefore a revised analytical weak-row formula or a richer nonlinear history source law that changes the eight closure rows themselves.

The next diagnostic looks at that weak-row formula locally. A one-step recurrent weak-row differential audit evaluates the current source-free recurrent weak closure in the smooth case at $h = 0.04$, using both the zero-initial and post-one-step history modes. It completes in 26.0 seconds, keeps rank 132, and has maximum residual 2.48×10^{-12} . However, the current recurrent weak

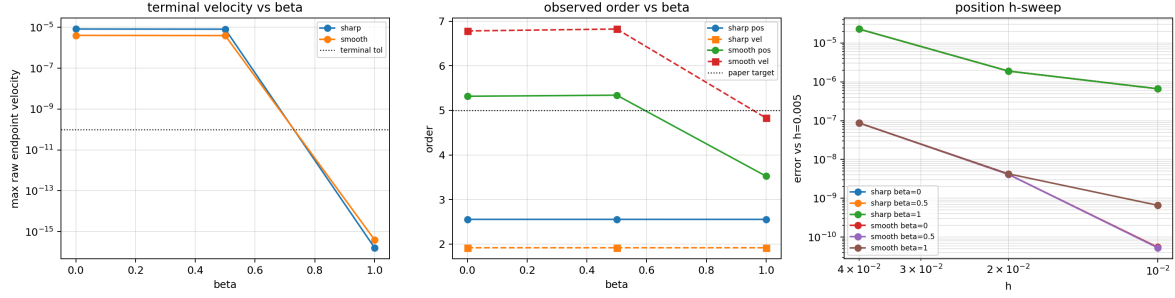


Figure 24: Order/closure blend trajectory audit. The beta path connects the order-preserving mean-acceleration closure to the terminal-closing final-stage velocity closure. It confirms that the two properties remain split.

row does not span the terminal-bridge tangent: the projection residual remains between 0.899 and 0.974 and the independent target rank remains eight. The final-stage terminal row, used only as a scoring comparison, projects at 1.46×10^{-15} . The missing direction is stage-2-local and dominated by translational velocity variables, while the relative recurrent-vs-final closure Jacobian gap reaches 6.872. This is the most concrete current blocker: the next formula must supply that stage-2 velocity coefficient-gradient contribution without substituting the final-stage terminal row.

A follow-up source/history coefficient-gradient probe tests exactly that direction without running a full trajectory solve. It modulates the stage-2 lower-pair velocity row by curvature-plus-history features with gains 10^8 and 10^{12} . The smooth local run covers both history modes and 8 candidate rows, completes in 35.3 seconds, keeps rank 132, and has maximum residual 2.48×10^{-12} . None of the candidates spans the terminal-bridge tangent: the best projection residual remains 0.899 and the independent target rank remains eight. The largest closure-value perturbation from the recurrent weak row is 1.25×10^{-9} , showing that this simple stage-2 coefficient-gradient correction perturbs the row value without adding the missing tangent. The next formula must therefore be richer than a scalar source/history modulation of the stage-2 velocity row.

A matrix-mixing follow-up makes that statement stronger while still avoiding a full trajectory solve. It applies target-free row-broadcast, column-broadcast, symmetric-broadcast, diagonal-plus-row-broadcast, shifted-column-broadcast, and bilinear stage-2 velocity coefficient matrices to the same stage-2 lower-pair velocity row, with gains 10^8 and 10^{12} . The smooth local run covers both history modes and 32 candidate rows, completes in 68.1 seconds, keeps rank 132, and has maximum residual 2.48×10^{-12} . None of these matrix-mixed rows spans the terminal-bridge tangent either: the best law is `diagonal_plus_row_broadcast_feature` at gain 10^{12} , with projection residual 0.898873 and independent target rank eight. Its largest closure-value perturbation from the recurrent weak row is 4.00×10^{-9} , so row/column mixing changes the value but still does not supply the missing stage-2 tangent. The remaining full-TFE repair target is therefore a genuinely different analytical weak-row formula, not a matrix premultiplication of the current stage-2 velocity row.

An active-velocity extension of the same matrix audit then uses the stage-2 velocity row itself, the stage-2 minus stage-1 velocity row, and the velocity curvature as coefficient features. This run again covers both history modes, 16 candidate rows, and gains 10^8 and 10^{12} ; it completes in 46.6 seconds with rank 132 and maximum residual 2.48×10^{-12} . The best law, `stage2_velocity_column_broadcast_feature` at gain 10^{12} , improves the local projection residual to 0.585746, but still has independent target rank eight and no terminal-bridge span. Its largest closure-value perturbation is 1.46×10^{-6} . Thus active stage-2 velocity dependence carries more of the missing tangent than

source/history features, but it still cannot be accepted as a full-TFE replacement or promoted to a trajectory h-sweep without a different weak-row formula.

A final small active-law grid widens that check without adding any new trajectory machinery. It tests symmetric, diagonal-plus-row, and shifted column broadcasts of the stage-2 velocity feature, plus column broadcasts of stage-2-minus-stage-1 and stage-0/1 mean velocity. The grid covers 20 local rows and completes in 51.2 seconds. The best row is `stage2_velocity_shifted_column_broadcast_feature` at gain 10^{12} , improving the projection residual only to 0.576548. The independent target rank remains eight and no row spans the terminal bridge. This leaves the active-velocity family useful as localization evidence, not as an acceptably derived eight-row lower-pair closure.

The best active-law row was then rerun alone with a missing-direction decomposition, again without invoking the full artifact pipeline. This targeted check covers 2 local rows in 27.8 seconds. The zero-initial row has projection residual 0.909693; its missing direction has rank eight, dominant variable family `translation_velocity_v` at fraction 0.487, and stage-2 fraction 0.965. The post-one-step row is the best row, with projection residual 0.576548; its missing direction again has rank eight, but the dominant variable family shifts to `angular_velocity_w` at fraction 0.503, with stage fractions 0.0185, 0.0185, and 0.963 for stages 0, 1, and 2. This makes the next repair target narrower: it should derive a stage-2 lower-pair weak row whose coefficient derivative couples the velocity closure to angular velocity, rather than continuing to widen generic matrix-premultiplication families.

A stage-2 angular-velocity mask follow-up tests the simplest version of that localization: keep only the angular components of the stage-2 lower-pair velocity feature. It covers 8 local rows in 35.3 seconds with four matrix laws and gain 10^{12} . The best law, `stage2_angular_velocity_shifted_column_broadcast_feature`, reaches projection residual 0.752997, worse than the unmasked active-law residual 0.576548. The independent target rank remains eight and no row spans the terminal bridge. The best missing direction shifts back to `translation_velocity_v` at fraction 0.660, with 97.8% of the energy still in stage 2. Thus the missing formula is not a simple angular-only mask; it likely needs coupled stage-2 translational/angular weak-row structure.

A direct translation/angular cross-coupling follow-up then tests the simplest outer-cross matrix versions of that idea. It covers 8 local rows in 35.3 seconds with four laws and gain 10^{12} . The best law, `stage2_velocity_angular_to_translation_cross_feature`, reaches projection residual 0.898812, again worse than the unmasked active-law residual 0.576548. The independent target rank remains eight and no row spans the terminal bridge. Its remaining direction is dominated by `translation_velocity_v` at fraction 0.556, with essentially all energy in stage 2. Thus simple outer-cross coefficient matrices are also not the missing weak row.

A target-free endpoint-pose velocity-predictor follow-up moves closer to the missing row family. It evaluates lower-pair velocity constraints at the paper terminal pose using several generalized-velocity predictors from the stage data. The best of 10 local rows uses the positive part of the Gauss-node terminal Lagrange velocity predictor. It reduces the projection residual to 0.117782, but still leaves independent target rank eight and no terminal-bridge span. The remaining direction is dominated by `translation_velocity_v` at fraction 0.514, with 96.2% of its energy in stage 0. This is the sharpest current local repair probe, but it remains diagnostic.

A Gauss endpoint-pose version of the same predictor family is negative. The best of 7 local rows is `gauss_endpoint_pose_positive_lagrange_z`; it reaches only residual 0.208599, leaves independent target rank eight and no span, and changes the dominant missing family to `lie_position_u` at fraction 0.651. Since this is worse than the paper endpoint-pose residual 0.117782, it is a ruled-out endpoint-pose alternative rather than a new repair direction. The remaining 11 Gauss endpoint-pose quarter, stage1-half, convex, and mean predictors are also negative. Their best row, `gauss_endpoint_pose_stage02_convex_0p00_z`, reaches residual 0.172659, leaves indepen-

dent target rank six and no span, and keeps the remaining direction dominated by `lie_position_u` at fraction 0.952340.

A near-terminal stage-0/stage-2 convex predictor screen clarifies the remaining boundary. The stage-2-only endpoint-pose row spans the terminal bridge at roundoff, but this is exactly the terminal-stage bridge row already known to be order-limited in nonlinear h-sweeps. The best nonterminal row uses a 5% stage-0, 95% stage-2 velocity predictor; it lowers the local residual to 0.049317 at $h = 0.04$, but still leaves independent target rank eight and no nonterminal span. Focused reruns at $h = 0.02$ and $h = 0.01$ give residuals 0.051890 and 0.052472. The terminal-equivalent row still spans at roundoff, but the nonterminal row does not converge to a span as h is reduced, so the remaining target is an analytical nonterminal weak row rather than a continuation toward the terminal bridge.

The synchronized pose/velocity version of the same test evaluates $C_v(q_\alpha, z_\alpha)$ at matching stage-0/stage-2 convex points. Its best row, `stage02_convex_pose_velocity_0p01_z`, reduces the local residual to 0.009512 at $h = 0.04$. However, it still leaves rank eight unspanned, and h-scaling gives 0.009978 at $h = 0.02$ and 0.010085 at $h = 0.01$. The residual is nearly proportional to the offset from the terminal row: 0p02 gives 0.019205, 0p05 gives 0.049390, and 0p10 gives 0.103464. This is useful localization evidence but still a near-terminal limit, not an independent full-TFE replacement.

A terminal-limit extrapolation check then combines two nonterminal synchronized rows. The 0p01/0p02 extrapolated law reduces the local residual to 4.699×10^{-6} at $h = 0.04$, and the wider 0p01/0p05 extrapolation gives 1.175×10^{-5} . Under h-scaling the 0p01/0p02 residual decreases to 2.335×10^{-6} and 1.168×10^{-6} at $h = 0.02$ and $h = 0.01$, but it still does not span the terminal bridge: the independent target ranks are 8, 8, and 7. We therefore mark these rows as terminal-bridge-equivalent extrapolations, not as independent full-TFE replacements.

Adding the centered bridge’s mean-acceleration term directly does not fix the remaining tangent. The signed acceleration-bridge variants worsen the projection residual from 4.699×10^{-6} to at least 0.076159 and move the dominant missing direction to `lie_position_u`. This excludes the direct mean-acceleration bridge term as the missing nonterminal correction.

A bilinear active-velocity outer-product follow-up then checks whether simple component-pair coupling between the coefficient feature and the stage-2 velocity row supplies the missing tangent. It covers 24 local rows in 57.3 seconds using feature/stage-2-velocity outer, reverse outer, symmetric outer, and related active-feature variants. The best row is `stage2_velocity_feature_outer_stage2_velocity` at gain 10^{12} , with projection residual 0.898720. The independent target rank remains eight and no row spans the terminal bridge. This rules out the tested bilinear outer-product coupling family; it does not improve on the previous shifted-column active-law localization residual 0.576548.

A componentwise active-velocity diagonal follow-up then tests whether the missing stage-2 tangent is row-local rather than an outer product. It covers 24 local rows in 56.7 seconds using Hadamard diagonal, shifted diagonal, and diagonal-plus-row/column variants. The best row is `stage2_velocity_diagonal_plus_hadamard_row_feature_stage2_velocity` at gain 10^{12} , with projection residual 0.898530. The independent target rank remains eight and no row spans the terminal bridge. This rules out the tested componentwise and shifted-diagonal active-velocity couplings as the missing eight-row weak closure.

Two acceleration-correction follow-ups test whether the synchronized near-terminal row only needs a simple Taylor term. The first adds the centered bridge mean-acceleration closure directly to `stage02_convex_pose_velocity_0p01_z` and its 0p02 analogue with signed coefficients ± 0.1 and ± 1 . The second applies the same signed coefficient as a generalized-velocity shift, $z_\alpha + \gamma h \dot{z}_\alpha$, before evaluating $C_v(q_\alpha, z_\alpha)$. Each grid covers 18 local rows, keeps rank 132, and records zero nonterminal

spans. In both grids the best row remains the uncorrected `stage02_convex_pose_velocity_0p01_z` with residual 0.009512; the best acceleration-corrected residual is only 0.012376, while unit corrections are no better than 0.077874. Thus a direct acceleration closure or generalized-velocity acceleration predictor is not the missing nonterminal full-TFE weak row.

A further geometry-oriented test replaces or perturbs z_α toward the stage-0/stage-2 pose slope $(q_2 - q_0)/(h(c_2 - c_0))$. This is meant to check whether the missing row is just a kinematic velocity estimate from the paper stage positions. It covers 26 local rows, again with no nonterminal span. The best row is still the uncorrected `stage02_convex_pose_velocity_0p01_z` at residual 0.009512; the best pose-slope row, `stage02_convex_pose_velocity_0p02_poseslope_m0p1_z`, has residual 0.878519 and leaves a stage-2 dominated translation-velocity direction. The tested pose-slope velocity predictor is therefore also not the missing full-TFE weak row.

Finally, a source-to-velocity lift test maps lower-pair source rows through the current endpoint velocity matrix $C_{v,z}(q_\alpha)$ into a minimum-norm generalized-velocity correction. This is a richer target-free source law than the scalar Taylor corrections: it uses the mean acceleration closure, stage source estimates, source curvature, and the history/source delta. The full grid covers 34 local rows in 76.4 seconds and again has zero nonterminal spans. The best row overall is still the uncorrected `stage02_convex_pose_velocity_0p01_z` at residual 0.009512. A focused rerun gives the best source-lift residual 0.609803 for `stage02_convex_pose_velocity_0p01_sourcelift_historydelta_p1_z` and the equivalent source0 branch. This excludes the tested current-pose C_v source-to-velocity lift as the independent weak row.

A final near-terminal slope probe tests the finite-difference slope of the same synchronized pose/velocity family rather than its terminal-limit intercept. It covers four local rows in 25.8 seconds, keeps rank 132, and again has no span. The best law, `stage02_convex_pose_velocity_slope_0p01_0p05_z`, reaches only projection residual 0.677034 with independent target rank eight and dominant remaining family `translation_velocity_v`. Thus the first nonterminal slope of the near-terminal convex row is also not the missing coefficient-gradient closure.

A near-terminal curvature probe then tests the second finite-difference curvature of the same synchronized pose/velocity family. It covers four local rows in 26.7 seconds, keeps rank 132, and again has no span. The best law, `stage02_convex_pose_velocity_curvature_0p01_0p05_0p10_z`, reaches only projection residual 0.871228 with independent target rank eight and dominant remaining family `translation_velocity_v`. Thus the local second derivative of the near-terminal convex row is also not the missing coefficient-gradient closure.

A source/history coefficient-feature matrix probe then tests recurrent features such as history delta, source0/source2, source02 delta, and source curvature as shifted-column coefficient matrices multiplying the stage-2 lower-pair velocity row. It covers 16 local rows in 44.8 seconds, keeps rank 132, and again has no span. The best law, `source_curvature_shifted_column_broadcast_feature`, reaches only projection residual 0.898873 with independent target rank eight and dominant remaining family `translation_velocity_v`. Thus the tested source/history coefficient features are also not the missing coefficient-gradient closure.

A nonlinear recurrent curvature/history matrix-feature extension then adds source-curvature minus history-delta, source/history Hadamard, unit-Hadamard, and normalized curvature-plus-history feature prefixes. It covers 16 local rows in 45.5 seconds, keeps rank 132, and again has no span. The best law, `source_curvature_minus_history_delta_diagonal_plus_row_broadcast_feature`, reaches only projection residual 0.898865 with independent target rank eight and a stage-2 dominated `translation_velocity_v` gap, with dominant fraction 0.556051 and stage-2 fraction 0.999996. Thus the tested nonlinear recurrent curvature/history matrix features are also excluded.

A terminal-limit source-lift matrix-difference probe then tests whether the best near-terminal extrapolated intercept can be repaired by adding a stage-local C_v source-to-velocity lift before ex-

trapolation. It covers eight local rows in 47.5 seconds, keeps rank 132, and again has no span. The best law is the `curvature_m0p1` source-lift variant; it reaches projection residual 5.298×10^{-6} with independent target rank eight and dominant remaining family `translation_acceleration_a`. This is close to, but not better than, the uncorrected terminal-limit residual 4.699×10^{-6} , so the tested C_v source-lift correction is also not the missing coefficient-gradient closure. A curvature source-lift scale refinement then tests the same terminal-limit row with signed curvature coefficients from 0.01 to 0.5. It covers eleven local rows in 52.9 seconds and again has no span; the best `curvature_p0p01` law still reaches only 5.298×10^{-6} , with independent target rank eight, dominant `translation_acceleration_a` fraction 0.407672, and stage fractions 0.549481/0.005742/0.444777. The local source-curvature norm is 1.076×10^{-14} , so this is not merely a missed source-lift scale. A mean-acceleration bridge scale refinement then tests the same terminal-limit row with signed bridge coefficients from 0.01 to 0.5. It covers eleven local rows in 31.4 seconds and again has no span. The best nonterminal law, `accel_p0p01`, reaches only 0.000867721 with independent target rank eight and dominant `lie_position_u` fraction 0.706665, while the overall best row remains the terminal-bridge-equivalent uncorrected extrapolation. Thus this is not merely a missed small acceleration-bridge coefficient. A generalized-acceleration Taylor-shift scale refinement then uses separate $h\dot{z}$ shifts at the left and right convex points before the same extrapolation. It covers eleven local rows in 30.2 seconds and again has no span. The best nonterminal law, `genaccel_p0p01`, reaches only 0.000866899 with independent target rank eight and dominant `lie_position_u` fraction 0.701135. Thus the missing row is not the tested stage-local generalized-acceleration Taylor shift. A pose-acceleration Taylor-shift scale refinement then moves the evaluated pose rather than only the velocity, using separate $h^2\ddot{z}$ shifts at the same left and right convex points. It covers eleven local rows in 29.9 seconds and again has no span. The best nonterminal law, `poseaccel_p0p01`, reaches 4.1835e-05 with independent target rank eight, dominant `lie_position_u` fraction 0.711570, and stage-2 fraction 0.771337. This pose-directed perturbation is closer than the velocity-only generalized-acceleration shift, but it still does not provide an independent full-TFE stage row. A tiny pose-acceleration scale refinement then lowers the signed coefficients to 0.0005, 0.001, 0.002, and 0.005, while retaining the previous positive 0.01 row. It covers ten local rows in 30.0 seconds and still has no span. The best nonterminal law, `poseaccel_p0p0005`, reaches 5.5898e-06 with independent target rank eight and dominant `translation_acceleration_a` fraction 0.350240. Thus small pose shifts only approach the terminal-equivalent row and do not produce an independent full-TFE closure row. A pose+velocity acceleration Taylor refinement then combines the small $h^2\ddot{z}$ pose shift with signed small $h\dot{z}$ velocity shifts. It covers eleven local rows in 30.7 seconds and again has no span. The best nonterminal law, `posevelaccel_p0p0005_m0p0001`, reaches 1.0173e-05 with independent target rank eight and dominant `lie_position_u` fraction 0.544660. Thus this simple combined Taylor correction is also not the missing independent full-TFE closure row. A component-split pose-acceleration refinement then separates the same $h^2\ddot{z}$ pose shift into translation-only and angular/Lie-only branches. The two JSON-only screens cover 26 local rows in 32.3+32.3 seconds, keep rank 132, and again have no span. Translation-only shifts through coefficient 0.1 are locally indistinguishable from the uncorrected terminal-limit row; the best nonterminal law, `transposeaccel_m0p01`, remains at 5.298e-06 with dominant `translation_acceleration_a` fraction 0.407672. The angular-only branch reproduces the earlier full pose-acceleration behavior: `angposeaccel_p0p002` reaches 9.6511e-06, and `angposeaccel_p0p01` reaches 4.1835e-05. Thus the observed pose-acceleration effect is angular/Lie-pose dominated; a translation-only pose shift is not the missing independent weak row.

A component-split velocity-acceleration Taylor refinement then performs the analogous split on the $h\dot{z}$ generalized-velocity shift. The first JSON-only screen covers 13 local rows in 31.8 seconds, keeps rank 132, and finds no span. Its best nonterminal row, `transvelaccel_m0p01`, reaches

projection residual 2.138×10^{-4} with independent target rank eight and dominant `translation_acceleration_a` fraction 0.886603. The best angular velocity branch, `angvelaccel_p0p01`, reaches only 0.000864560 and moves the dominant missing family to `lie_position_u`. A tiny translational scale follow-up covers 11 local rows in 31.4 seconds; its best law, `transvelaccel_m0p0005`, reaches 9.8175×10^{-6} , still worse than the uncorrected terminal-limit residual 5.298×10^{-6} . Thus the velocity-acceleration split only approaches the terminal-equivalent limit and does not provide an independent full-TFE closure row.

A component-mixed pose/velocity Taylor refinement then combines the two most promising split directions: angular/Lie pose shift with translational velocity shift, plus the complementary translation-pose/angular-velocity branch. It covers 13 local rows in 32.0 seconds, keeps rank 132, and again finds no span. The best nonterminal row is the `p0.0005/m0.0005` angular-pose/translational-velocity row. It reaches only 9.987×10^{-6} , with independent target rank eight and a dominant translation-acceleration fraction of 0.774582. The complementary translation-pose/angular-velocity branch reaches only 0.000864560. This is still worse than the uncorrected terminal-limit residual 5.298×10^{-6} , so the tested component-mixed Taylor cross term is also not the missing independent full-TFE closure row.

A stage-2-fixed/delta acceleration velocity-shift refinement then tests whether the dominant translation-acceleration residual is an artifact of using the interpolated acceleration source. It covers 13 local rows in 31.8 seconds, keeps rank 132, and again finds no span. The best nonterminal row is `transvelaccelstage2_m0p0005`, with residual 9.8175×10^{-6} and translation-acceleration fraction 0.802349. The best delta20 row reaches only 1.0119×10^{-5} . Thus stage-2-fixed and delta20 acceleration sources reproduce the earlier tiny-shift limit rather than providing the independent full-TFE closure row.

A nonfinal terminal velocity/source predictor refinement then tries to predict the terminal lower-pair velocity closure from only stage-0/stage-1 velocity rows and source estimates. It covers nine local rows in 30.9 seconds, keeps rank 132, and finds no span. The best law, `euler1`, reaches only 0.923466, with dominant `translation_velocity_v` fraction 0.526951 and stage-2 fraction 0.959959. Thus a plain nonfinal velocity/source integral predictor does not supply the missing stage-2 terminal-bridge Jacobian direction. The corresponding bounded trajectory smoke confirms this at nonlinear-solve level: the stage-0/stage-1 family converges 24/24 rows with rank 132, but its best terminal velocity is only 1.770×10^{-7} , and the order-friendlier `euler1/ab01/source01linear1` rows remain at 3.584×10^{-7} . A stage-0/1/2 extension using the third Gauss stage also converges all 18 rows with `projection_used=false`, but `terminal_velocity_closed=false`: the closest rows reach 2.559×10^{-11} at $h = 0.04$ and then jump to 1.328×10^{-7} at $h = 0.02$, while the stable `linear012` row is only 1.134×10^{-7} .

A stage-2 source-to-velocity transport refinement applies the source/history projection directly at the stage-2 pose/velocity closure. The first screen covers 13 local rows in 53.4 seconds and reaches 0.001251 with independent target rank six. A scale refinement over the matrix-difference history-delta family covers 11 rows in 57.7 seconds and improves the best residual to 0.0001251 at `historydelta_m0p01`. An ultra-fine scale follow-up covers nine rows in 49.6 seconds and pushes the best nonterminal residual to 1.251×10^{-6} at `historydelta_m0p0001`. This is the strongest nonterminal transport clue so far, but it still has no span and leaves a rank-six `lie_position_u` missing direction. A combined source-transport/angular-pose follow-up covers eight rows in 48.7 seconds; the source-only row remains best, and the best combined row reaches only 1.296×10^{-6} while raising the independent target rank to eight and still finding no span. Promoting those two source-transport rows into the bounded nonlinear trajectory smoke does not change the acceptance boundary. The three-law comparison runs in 74.1 seconds, keeps rank 132, uses no projection, and writes no artifacts. The source-only transported row `stage02_convex_pose_velocity_`

`Op00_sourceliftmatrixdiff_historydelta_m0p00001_z` keeps the two-point trajectory orders at 6.588/4.508, but its max terminal velocity is 8.020×10^{-12} , above the 10^{-12} closure tolerance. Adding the small angular-pose shift worsens the terminal velocity to 2.102×10^{-7} and lowers the velocity order to 2.820. Thus the source-to-velocity transport clue is now a trajectory-level negative as well as a local no-span result. A three- h refinement of the source-only transported row confirms that this is not a two-point artifact: it converges all seven short-sweep steps in 46.4 seconds, keeps rank 132 and no projection, but its maximum terminal velocity is still 8.020×10^{-12} even though the finest- h value is 5.904×10^{-13} . Its position/velocity orders are again only 4.142/2.305, matching the bare endpoint-row order loss rather than recovering the full-TFE acceptance target. Shrinking the same source-transport coefficient by another factor of ten gives a useful limiting negative: the `stage02_convex_pose_velocity_Op00_sourceliftmatrixdiff_historydelta_m0p00001_z` row closes the terminal tolerance with max/finest terminal velocity $8.021 \times 10^{-13}/5.910 \times 10^{-14}$, but the smooth orders remain 4.142/2.305. Coefficient shrinking can therefore recover terminal closure in this family, but only by approaching the same order-limited endpoint-row behavior.

8 Next Full-TFE Repair Target

The remaining full-TFE work is now narrow enough to state as a concrete repair target. It is not another four-example validation run, and it is not a request to rerun the complete historical audit harness. The missing object is an eight-row, source-free lower-pair closure formula that can replace the terminal bridge while preserving the trajectory order of the paper-derived stage residual.

This target also explains why the recent negative audits are useful. They do not show that TFE is impossible; they show which tempting shortcuts are not the missing row. Frozen row-space compression can balance values without closing the tangent gap. Near-terminal convex rows can approach the terminal bridge but remain terminal-equivalent. Current-pose and stage-difference C_v source-to-velocity lifts move source rows into generalized velocity space, but still do not span the eight missing directions. The next useful implementation therefore has to change the weak-row formula itself, most likely through a genuinely nonlinear recurrent history source law or a stage-2 translational/angular coefficient-gradient term derived from the lower-pair weak balance.

9 Sparse Backend Status

The row-colored VJP backend uses 60 row colors rather than the 90 column colors of the dense-pattern coloring. It preserves trajectory agreement, but the current implementation is not a consistent runtime win over dense `jacfwd`. The sparse cost-model audit records the required row-runtime reduction needed to match dense and to achieve a 10% dense win. This is an engineering caveat, not a correctness caveat.

10 Reproducibility Artifacts

The primary generated artifacts are:

- `v047_cylindrical_chain_pipeline/run_v047.py`;
- `v047_cylindrical_chain_pipeline/validate_v047_outputs.py`;
- `v047_cylindrical_chain_pipeline/validate_full_tfe_gap.py`;

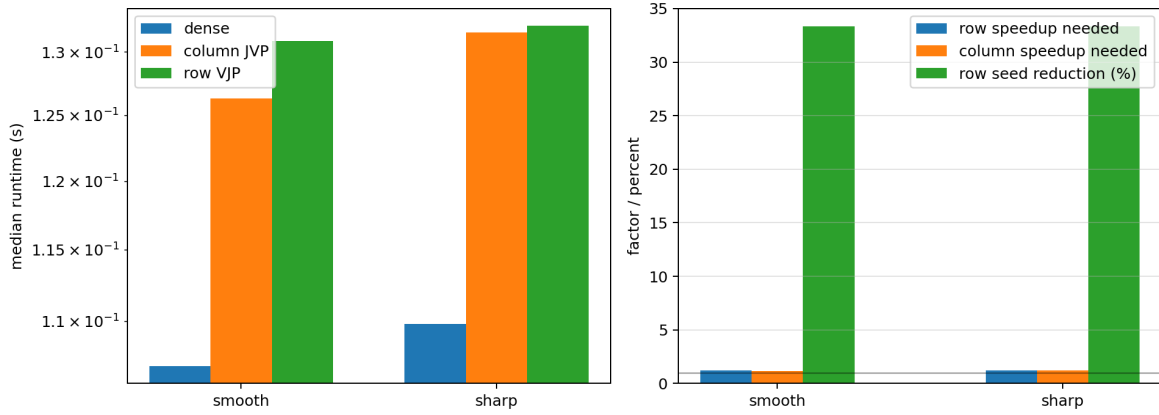


Figure 25: Sparse speed gap audit. Row-VJP reduces the seed count but is not yet a stable wall-clock improvement over dense `jacfd`.

- `v047_cylindrical_chain_pipeline/validate_full_tfe_repair_spec.py`;
- `v047_cylindrical_chain_pipeline/FULL_TFE_REPLACEMENT_GAP_LEDGER.md`;
- `v047_cylindrical_chain_pipeline/FULL_TFE_REPAIR_SPEC.md`;
- `v047_cylindrical_chain_pipeline/results/summary_v047.json`;
- `v047_cylindrical_chain_pipeline/results/v047_report.md`;
- `ORDER_PROOF_LEDGER.md`;
- `VALIDATION_QUICKSTART.md`;
- `paper_v047_cylindrical_chain/CLAIM_BOUNDARY.json`;
- `paper_v047_cylindrical_chain/REVIEWER_CHECKLIST.md`;
- the CSV and PNG files under `v047_cylindrical_chain_pipeline/results/`.

The machine-readable claim boundary is `CLAIM_BOUNDARY.json` in this paper directory. It records the accepted formal-order comparison, the local paper-style comparator, the explicit non-claims, the open caveats, and an `evidence_files` manifest. That manifest points to the summary JSON, convergence CSV, ASME CSV rows, paper-TFE formula-mapping CSV, full-TFE gap ledger, repair specification, paper PDF, claim ledger, reviewer checklist, and Chinese status note. The paper-claim validator parses this JSON and checks those evidence paths before accepting the paper package.

At the time of this build, the validator expects the source-free closure, the order/closure blend, and the velocity-compression probe artifacts through the row-space compression, coefficient-derivative, state-feature coefficient-derivative, and four-history source-law audits plus the JSON-only active-stage velocity matrix-gradient localization. The four-history source-law screen is negative evidence: it keeps `terminal_closed_row_count=0` while the best smooth-order branch has min order 5.342. The rebuilt v047 artifact set passes the v047 validator with 292 result files, 147 CSV files, and 120 PNG files; the top-level pipeline validator also passes with 48 version directories, 695 result files, 286 CSV files, 202 PNG files, and 113 JSON files.

The full numerical regeneration is deliberately not the default paper build path. The latest recorded full v047 generation time is 2386.76s, because `run_v047.py` has accumulated the complete historical audit suite. For the four-example paper claim, the authoritative evidence is the existing generated artifact set plus the read-only validators. A new full regeneration is needed only when numerical CSV/JSON/PNG evidence changes. Accordingly, `run_v047.py` should be read as an audit harness and artifact generator, not as a compact reference implementation of the accepted integrator. Its size reflects the accumulated negative full-TFE screens and historical diagnostics. It is not necessary to execute that harness merely to check the four-example claim or rebuild this paper. The paper build and paper-claim check run from this paper directory; the v047 artifact checks run from the v047 pipeline directory. The reviewer checklist in this directory gives the shortest read-only command sequence for checking the accepted method claims, the diagnostic full-TFE boundary, and the explicit rule that `run_v047.py` should not be run for routine paper edits. For full-TFE blocker exploration, the preferred workflow is a targeted `V047.TARGET_AUDIT` run that prints JSON for one local audit. The recent active-stage velocity matrix grid used this path, took 51.2 seconds, and did not regenerate the full artifact set. Later terminal-limit Taylor-style probes used the same JSON-only pattern: an earlier pose+velocity acceleration refinement covered 11 local rows in 30.7 seconds, found best nonterminal row `stage02_convex_pose_velocity_extrapolate_0p01_0p02_posevelaccel_p0p0005_m0p0001_z` at residual 1.0173×10^{-5} , and still found no terminal-bridge span. The latest stage-2 source-to-velocity transport screen is the strongest nonterminal transport clue so far: the scaled `stage02_convex_pose_velocity_0p00_sourceliftmatrixdiff_historydelta_m0p0001_z` row reaches residual 1.251×10^{-6} , reduces the independent target rank to six, and still has no span. Adding a small angular/Lie pose shift does not improve it: the best combined row reaches only 1.296×10^{-6} and also has no span. A final three-row bare endpoint-pose velocity check finds that `stage02_convex_pose_velocity_0p00_z` spans the local terminal bridge at roundoff, with residual 1.327×10^{-15} , but this is recorded as a local row-space span rather than an accepted full-TFE replacement: no trajectory h-sweep, order proof, or artifact update is attached. These probes are intentionally diagnostic local evidence; they do not update the accepted four-example artifact set. A bounded JSON-only trajectory smoke now promotes that local span into the nonlinear 132-row residual. The `stage02_convex_pose_velocity_0p00_z` row closes terminal velocity to 8.124×10^{-17} on a smooth three-*h* short sweep, but its position/velocity orders are only 4.142/2.305. The nearby nonterminal `stage02_convex_pose_velocity_0p01_z` row converges but leaves terminal velocity at 6.255×10^{-6} . This is trajectory-level diagnostic evidence, not an accepted full-TFE replacement. A non-projection extrapolation limit reaches terminal velocity 1.254×10^{-13} at `stage02_convex_pose_velocity_extrapolate_0p00001_0p00002_z`, but the three-*h* order remains 4.142/2.305, so the terminal/order split is not removed by taking the nonterminal points closer to the endpoint.

11 Non-Claims and Limitations

The accepted evidence is intentionally narrower than the complete research program. The paper therefore makes the following non-claims explicit.

12 Remaining Gaps

13 Conclusion

The v047 evidence supports the intended conditional formal-order comparison. The accepted Gauss6/FullVA path follows the conditional sixth-order Gauss/FullVA theorem, records smooth position/velocity orders 7.161/7.066 on the cylindrical lower-pair benchmark, and has four-example coverage with dynamic-order evidence on the single- and double-pendulum rows and closed-loop residual/reaction evidence on the four-link and slider-crank rows. In the current artifact scope, this is the result to compare against the local paper-style $m = 3$ Lobatto-TFE formula target with expected order five. The sharp-friction behavior is localized to a practical coarse-regime issue with high-order recovery under deeper fixed refinement, not to a failure of the smooth conditional sixth-order method claim.

The independent full-TFE replacement remains open. The source-free final-stage velocity closure closes raw terminal endpoint velocity without projection, endpoint source data, or terminal-row replacement, but its smooth position order is too low for an accepted paper-TFE replacement. The newer endpoint-pose velocity predictor smoke shows the same split at trajectory level: the terminal-equivalent row closes velocity but loses order, while the nearby nonterminal row does not close terminal velocity. Ultra-near non-projection extrapolation closes the terminal velocity tolerance at 1.254×10^{-13} but retains the same 4.142/2.305 order loss; a three-point quadratic nonterminal extrapolation similarly closes at 2.029×10^{-13} with finest- h terminal velocity 8.254×10^{-15} and the same order loss. The machine-readable gate remains `full_tfe_stage_replacement=false`. The latest compression diagnostics rule out the tested target-free linear direction laws, finite-difference second-differential non-final source features, higher-order/nonlocal source features, one-step and two-step history-transport features, recurrent curvature/bilinear history features, source-active cross-Gram/Hadamard/shifted-commutator weak-row structures, and target-free frozen row-space compressions. The latest active-stage velocity matrix-gradient localization improves the residual to 0.576548 but still has rank-eight missing directions and no terminal-bridge span. The best-row decomposition puts 96.3% of the remaining tangent energy in stage 2 and identifies `angular_velocity_w` as the dominant variable family. A simple angular-only mask follow-up reaches only 0.752997 and shifts the remaining direction back to `translation_velocity_v`; a direct translation/angular cross-coupling follow-up reaches only 0.898812; an endpoint-pose positive-Lagrange velocity predictor improves the local residual to 0.117782 but still has rank eight/no span; a near-terminal convex predictor has a terminal-equivalent local span but its best nonterminal row remains rank eight/no span at 0.049317, 0.051890, and 0.052472 over $h = 0.04, 0.02, 0.01$; synchronized pose/velocity near-terminal rows reduce the best local residual to 0.009512 but keep rank eight/no span; nonterminal terminal-limit extrapolation reduces the residual to 4.699×10^{-6} but still has no span and is marked terminal-bridge-equivalent; direct near-terminal slope rows reach only 0.677034; direct mean-acceleration bridge correction worsens the residual to 0.076159 or above; a direct near-terminal curvature screen reaches only 0.871228; a source/history coefficient-feature matrix screen reaches only 0.898873; a nonlinear recurrent curvature/history matrix-feature screen reaches only 0.898865; a terminal-limit C_v source-lift screen reaches only 5.298×10^{-6} ; a component-split velocity-acceleration Taylor screen reaches only 2.138×10^{-4} at ordinary scale and 9.8175×10^{-6} at tiny translational scale; a component-mixed pose/velocity Taylor screen reaches only 9.987×10^{-6} ; a stage-2-fixed/delta acceleration velocity-shift screen reaches only 9.8175×10^{-6} ; a nonfinal terminal velocity/source predictor screen reaches only 0.923466; a stage-2 source-to-velocity transport screen reaches 1.251×10^{-6} but no span, and an angular-pose combined follow-up reaches only 1.296×10^{-6} with no span; the bare `stage02_convex_pose_velocity_0p00_z` row spans locally

at 1.327×10^{-15} and closes terminal velocity on a bounded trajectory smoke, but the three- h smooth orders are only 4.142/2.305; the nonterminal *0p01* trajectory row leaves terminal velocity at 6.255×10^{-6} ; a Gauss endpoint-pose predictor screen reaches only 0.172659 and shifts the remaining family to `lie_position_u`; a bilinear outer-product follow-up reaches only 0.898720; and a componentwise diagonal follow-up reaches only 0.898530. All likewise have no span. The next technical target is a revised analytical weak-row formula or richer nonlinear recurrent history source law inserted into the nonlinear lower-pair h -sweep, preserving terminal closure while restoring the expected paper-TFE order.

Table 2: Current claim boundary. The order labels are intentionally separated because they refer to different residuals.

Claim	Status	Evidence or blocker
Formal-order comparison relative to the local paper-style TFE target	Accepted bounded comparison	Gauss6/FullVA has a conditional order-six theorem by the direct residual-bridge proof and records smooth observed orders 7.161/7.066; the encoded $m = 3$ Gauss–Lobatto TFE formula target has expected order five. This is not a full residual reproduction claim.
Gauss6/FullVA cylindrical-chain method	Accepted method path	Direct 132-row bridge: 96 non-dynamic rows supply row-local $O(h^7)$ residual input and 36 Newton–Euler rows vanish by direct substitution on the lifted Gauss stage, giving the conditional local $O(h^7)$ /global $O(h^6)$ claim under the stated interfaces.
Four ASME-style method rows	Accepted method gate	Single and double pendulum method-side FullVA rows are accepted; four-link and slider-crank closed-loop kinematic FullVA plus reaction-dynamics rows are accepted.
Paper formula mapping	Mapped diagnostic	The encoded $m = 3$ Gauss–Lobatto TFE operator has expected order five, but mapping the formulas is not the same as solving a full replacement residual.
Source-free terminal lower-pair closure	Useful but not accepted as full TFE	Raw terminal endpoint velocity closes to 4.01×10^{-16} , but smooth position order is 3.523.
Independent full-TFE stage replacement	Open	The missing eight source-free lower-pair closure rows must preserve terminal closure and restore the paper-TFE h-sweep order without endpoint source, projection, terminal-row replacement, or a target-direction oracle.

Table 3: Comparator definition for the paper-facing formal-order comparison.

Object	Role in this paper	Current evidence boundary
Accepted Gauss6/FullVA path	Method being claimed	Gauss/FullVA nonlinear residual with a conditional order-six theorem by direct 132-row residual bridge; smooth observed orders 7.161/7.066; four-example evidence separated into dynamic-order pendulum rows and residual/reaction coverage on four-link/slider crank.
Local paper-style $m = 3$ Lobatto-TFE target	Comparator used for Proposition 1	Formula-mapping target with expected order five. It is sufficient for the current formal-order comparison because the comparison is an order and evidence-scope claim.
Complete source-paper TFE residual	Not used as the accepted comparator	The source paper has a full TFE formulation, but v047 has not yet accepted an implementation in which all paper-derived stage rows replace the Gauss6/FullVA rows inside Newton. This is the open <code>full_tfe_stage_replacement=false</code> reproduction gate.
v046/rA baseline	Historical reproducibility anchor	Useful for repository regression context, but not the main paper-facing comparator for Proposition 1.

Table 4: Original-paper boundary versus accepted v047 claim.

Question	Source-paper/local target	Accepted v047 claim
What is being compared?	The local paper-style $m = 3$ Lobatto-TFE formula target, whose expected order is five in the formula-mapping audit.	The accepted Gauss6/FullVA path, a conditional order-six Gauss/FullVA one-step map with smooth observed orders 7.161/7.066.
What is implemented as the method?	The complete source-paper TFE residual is not used as the accepted comparator because the independent replacement gate remains open.	The nonlinear solve uses the accepted 132-row Gauss/FullVA residual and the audited projection/KKT endpoint closure.
What is proved by the artifacts?	They prove the local order-five paper-style target boundary, not a completed source-paper residual reproduction.	They support a paper-facing formal-order comparison inside the current v047 artifact scope and separated four-example ASME evidence gate.
What remains outside the claim?	Full source-paper residual reproduction requires accepted paper-derived stage rows inside Newton.	<code>full_tfe_stage_replacement=false</code> remains an explicit caveat and future stronger gate.

Table 5: Acceptance criteria for the paper-facing formal-order comparison.

Criterion	Required evidence	Status
Accepted method path	The claimed method is the Gauss6/FullVA one-step map in Section 2.1, not a diagnostic paper-TFE substitution.	Pass
Order exceedance	The direct residual-bridge theorem gives order six under the stated interfaces, and the accepted smooth h-sweep records position/velocity orders 7.161/7.066, above the encoded local $m = 3$ Lobatto-TFE expected order-five target.	Pass
Four-example gate	The accepted method rows cover single pendulum, double pendulum, four link, and slider crank under the same read-only ASME gate.	Pass
Comparator boundary	The comparison is against the local paper-style formula target; complete source-paper residual reproduction remains a separate gate.	Pass
Caveat separation	Sparse speed, sharp-friction coarse-regime behavior, and <code>full_tfe_stage_replacement=false</code> are explicit caveats and do not contradict the narrower comparative claim.	Pass
Read-only reproducibility	The paper-claim, paper-package, four-ASME, full-TFE gap, repair-spec, and pipeline-output validators check the claim without invoking the full v047 generator.	Pass

Table 6: Status checklist for interpreting the v047 claims.

Question	Answer	Evidence boundary
Is the main formal-order comparison accepted?	Yes	Gauss6/FullVA is the accepted conditional order-six method path; the smooth h-sweep records 7.161/7.066, above the local paper-style $m = 3$ Lobatto-TFE expected order-five target.
Does the four-example method-evidence gate pass?	Yes	The accepted method gate covers single pendulum and double pendulum as dynamic-order evidence, and four link and slider crank as lower-pair constraint/reaction coverage on the Gauss6/FullVA path.
Is the production method sixth order?	Conditionally	The theorem proves local $O(h^7)$ /global $O(h^6)$ on the selected smooth branch under the compact-tube, endpoint, implementation-binding, and solver-scale interfaces; the smooth h-sweep records observed orders 7.161/7.066.
Is the paper $m = 3$ Lobatto-TFE formula mapped?	Yes	The formula-mapping audit records expected order five, but this is not by itself a solved method replacement.
Is the independent full-TFE stage replacement accepted?	No	The source-free terminal closure closes endpoint velocity, but the replacement candidate remains order-limited and the machine-readable gate remains <code>full_tfe_stage_replacement=false</code> .

Table 7: Order conventions used in the paper. The rows are deliberately separated because they refer to different nonlinear residuals.

Path	Order statement	Interpretation
Accepted Gauss6/FullVA path	Conditional order-six method; observed smooth orders 7.161/7.066	This is the method used for the accepted cylindrical-chain and four-example evidence claims.
Paper $m = 3$ Lobatto-TFE formula map	Expected order five	This is a formula-mapping diagnostic. It becomes a method claim only after the paper-derived rows replace the stage residual inside Newton.
Current source-free terminal-closed TFE candidate	Smooth orders 3.523/4.828	Terminal velocity is closed without endpoint source, terminal-row replacement, or projection, but order is too low for an accepted full-TFE replacement.
Endpoint-pose velocity predictor smoke	Smooth orders 4.142/2.305	The terminal-equivalent 0p00 row closes terminal velocity to 8.124×10^{-17} , but the trajectory order remains too low; the nearby nonterminal 0p01 row leaves terminal velocity at 6.255×10^{-6} .

Table 8: Proof/status boundary inherited from the order-proof ledger.

Object	Status	Reason
Gauss6/FullVA smooth cylindrical method	Conditional order-six argument	Direct 132-row bridge gives local $O(h^7)$ residual defect and global $O(h^6)$ grid error under the retained compact-branch, endpoint, and solver-scale interfaces.
Sparse Jacobian backend	Order-preserving backend change	Exact AD and sparse linear algebra evaluate the same residual/Jacobian to the same Newton tolerance.
Endpoint closure path	Diagnostic caveat	Raw endpoint/projection residuals remain diagnostics; the theorem endpoint object is the velocity-level/KKT closure functional, so projection monitors are not used to claim independent full-TFE acceptance.
Paper $m = 3$ Lobatto-TFE mapping	Mapped but not accepted as a method	Expected order five is recorded for the formula map, but the rows have not yet produced an accepted full replacement h-sweep.
Independent full-TFE replacement	Open proof obligation	The source-free terminal closure is algebraically useful, but the current h-sweep remains order-limited and the gate is still <code>full_tfe_stage_replacement=false</code> .

Table 9: Traceability for the formal-order comparison proposition.

Proposition component	Artifact evidence	Read-only check
Accepted Gauss6/FullVA one-step map	Summary JSON, convergence CSV, and Section 2.1	Paper-claim validator
Smooth high-order behavior	Convergence CSV and convergence figure record 7.161/7.066 orders	v047 and paper-package validators
Four-example acceptance	Summary JSON ASME gate with single pendulum, double pendulum, four link, and slider crank	Minimal four-ASME validator
Comparator boundary	Comparator table and paper claim ledger separate the local $m = 3$ Lobatto-TFE target from complete source-paper residual reproduction	Paper-claim validator
Open full-TFE caveat	Full-TFE gap ledger and repair specification keep <code>full_tfe_stage_replacement=false</code> explicit	Full-TFE gap and repair-spec validators
Corollary acceptance	Assumption-lemma-theorem-corollary chain accepts the proposition only inside the current read-only v047 artifact scope	Paper-claim validator
Repository synchronization	Order-proof ledger, pipeline audit, and version ledger carry the same claim boundary	Pipeline-output validator

Table 10: Cylindrical-chain observed orders for the accepted Gauss6/FullVA path.

Case	Step sizes	Position order	Velocity order
Smooth, projected velocity	0.04, 0.02, 0.01	7.161	7.066
Smooth, raw endpoint	0.04, 0.02, 0.01	7.161	7.066
Sharp, projected velocity	0.04, 0.02, 0.01	1.894	2.683
Sharp, raw endpoint	0.04, 0.02, 0.01	1.894	2.683

Table 11: Sharp-friction refinement evidence.

Audit	Step window	Position order	Velocity order
Fixed refinement	0.02, 0.01, 0.005	3.294	5.867
Deep fixed refinement	0.01, 0.005, 0.0025	6.398	5.870
Ultra fixed refinement	0.005, 0.0025, 0.00125	7.521	5.725

Table 12: Four-example evidence summary.

Example	Accepted evidence
Single pendulum	Exact driven kinematic mapping; full absolute-coordinate driven FullVA residual; embedded scalar FullVA bridge; reaction-multiplier reconstruction.
Double pendulum	Method-side FullVA mapping with accepted nested local v029/FullVA self-reference; v046 retained as reproducibility and reference-floor diagnostic.
Four link	Lower-pair graph bridge, full-row-rank Φ_q audit, accepted local closed-loop kinematic FullVA solve, and accepted reaction-dynamics rows.
Slider crank	Lower-pair graph bridge, full-row-rank Φ_q audit, accepted local closed-loop kinematic FullVA solve, and accepted reaction-dynamics rows.

Table 13: Quantitative evidence behind the accepted four-example gate.

Example	Quantitative evidence	Artifact path
Single pendulum	Absolute-coordinate driven FullVA residual has minimum observed order 6.024 across position, velocity, orientation, and angular velocity; max stage residual is 3.664×10^{-12} .	Single absolute FullVA runs CSV
Double pendulum	Accepted local v029/FullVA self-reference has minimum orientation/angular velocity order 6.089; endpoint position/velocity constraint norms stay below 7.034×10^{-13} and 8.323×10^{-12} .	Double method runs CSV
Four link	Closed-loop kinematic FullVA solve over $h = \{0.02, 0.01, 0.005\}$ keeps position/velocity/acceleration constraint norms below 1.052×10^{-14} ; reaction-dynamics residual is at most 1.338×10^{-13} .	Closed-loop kinematic and reaction CSVs
Slider crank	Closed-loop kinematic FullVA solve over $h = \{0.02, 0.01, 0.005\}$ keeps position/velocity/acceleration constraint norms below 1.204×10^{-15} ; reaction-dynamics residual is at most 6.492×10^{-15} .	Closed-loop kinematic and reaction CSVs

Table 14: Representative paper-TFE replacement audits. “Full” denotes accepted independent full-stage replacement, not merely a successful local solve.

Audit	Max residual	Smooth orders	Full TFE?
Position-row substitution	5.68×10^{-12}	1.227 / 0.731	No
Kinematic-row substitution	9.57×10^{-12}	1.406 / 1.542	No
All-row substitution	9.64×10^{-12}	1.406 / 1.542	No
Consistent-z0 terminal output	9.79×10^{-12}	4.046 / 4.420	No
Lower-pair acceleration terminal output	8.35×10^{-12}	4.937 / 4.799	No
Source-free mean-velocity closure	8.13×10^{-12}	3.512 / 4.448	No
Source-free final-stage velocity closure	9.49×10^{-12}	3.523 / 4.828	No
Source-free order/closure blend, best-order beta	8.90×10^{-12}	5.341 / 6.825	No
Endpoint-pose predictor, 0p00 three- <i>h</i> smoke	9.76×10^{-12}	4.142 / 2.305	No

Table 15: Acceptance test for a future independent full-TFE replacement.

Requirement	Acceptance evidence
Stage residual replacement	The nonlinear solve uses the paper-derived 132 stage rows, including the new eight lower-pair closure rows, rather than appending an endpoint correction to a Gauss trajectory.
Source-free construction	The lower-pair closure uses only stage-local state/history data; it does not use endpoint-boundary source data, terminal-row replacement, projection, or a target-direction oracle.
Algebraic health	The solve keeps rank 132, residuals near the existing Newton tolerance, and well-recorded conditioning diagnostics on the same smooth/sharp cases.
Trajectory evidence	The candidate is promoted from local tangent probes to a nonlinear h-sweep over $h = \{0.04, 0.02, 0.01\}$ against the $h = 0.005$ reference, with CSV/JSON/plot artifacts.
Order and closure	The h-sweep must close raw terminal endpoint velocity while restoring the paper-TFE order target; the current source-free terminal-closed candidate fails here because its smooth position order is 3.523.
Pipeline synchronization	The summary JSON, full-TFE gap ledger, repair spec, paper claim ledger, and read-only validators must all move from the current <code>full_tfe_stage_replacement=false</code> boundary to the new accepted claim.

Table 16: Statements that are not claimed by the current v047 evidence.

Non-claim	Current reason
The paper-TFE method is fully accepted	False. The machine-readable gate remains <code>full_tfe_stage_replacement=false</code> ; the accepted trajectory method is the sixth-order Gauss6/FullVA path.
The four-example gate requires a new full regeneration	False. The four-example coverage rows are already accepted in the existing artifact set and are checked by the lightweight four-example validator, but the four-link and slider-crank rows are not dynamic-order proof rows.
<code>run_v047.py</code> is a compact reference implementation	False. It is an accumulated audit harness and artifact generator containing historical positive and negative screens.
Sparse AD is a speed win	Not yet. Row-VJP reduces seed count but is not a stable wall-clock improvement over dense <code>jacfwd</code> .
Sharp Brown–McPhee friction is inexpensive at coarse steps	Not yet. Ultra refinement recovers high order, but the practical coarse regime remains order-reduced and more expensive.

Table 17: Recommended reproducibility levels for this paper build.

Level	Command or artifact	Purpose
Paper build	<code>cmd.exe /c latexmk -pdf -interaction=nonstopmode main.tex</code>	Rebuilds the PDF from the current text and figure snapshot.
Paper package check	<code>validate_paper_package.py</code>	Runs the lightweight paper/four-example/artifact checks against the existing PDF; pass <code>--latex</code> to attempt a rebuild from the wrapper.
Paper-claim check	<code>validate_paper_claims.py</code>	Checks that this paper’s headline claims match the current summary, figures, and PDF.
Four-example gate check	<code>validate_four_asme_minimal.py</code>	Checks only the existing single, double, four-link, and slider-crank method evidence.
Full-TFE gap check	<code>validate_full_tfe_gap.py</code>	Checks the open replacement boundary and verifies that terminal closure, order/closure split, and compression blockers remain explicit.
Full-TFE repair spec check	<code>validate_full_tfe_repair_spec.py</code>	Checks the next candidate’s 132-row residual budget and the 16+8 lower-pair implementation contract, including the bounded smoke entry.
Read-only artifact check	<code>validate_v047_outputs.py</code>	Checks the already-generated v047 CSV/JSON/PNG/report evidence without rerunning the simulations.
Full numerical regeneration	<code>run_v047.py</code>	Regenerates the full historical v047 audit suite; use this only after changing the numerical method or artifact-producing code.

Table 18: Current missing items after the accepted four-example method gate.

Gap		Current status
Independent stage replacement	full-TFE	Open. Source-free terminal closure is available, but the accepted-order nonlinear h-sweep is still missing. The latest 36 target-free compression rows, velocity-matrix, source/history, terminal-limit source-lift, curvature source-lift scale, acceleration-bridge scale, generalized-acceleration shift, pose-acceleration shift, tiny pose-acceleration scale, pose+velocity Taylor shift, component-split pose-acceleration, component-split velocity-acceleration, component-mixed pose/velocity Taylor, and Gauss endpoint-pose probes all have zero spans. The closest terminal-limit source-lift family remains at 5.298×10^{-6} with source-curvature norm 1.076×10^{-14} , so the next repair cannot be a simple source-lift scale tweak. The strongest nonterminal transport clue is the stage-2 source-to-velocity transport screen: it reaches residual 1.251×10^{-6} and lowers the independent target rank to six, but it still has no span; the angular-pose combined follow-up reaches only 1.296×10^{-6} and remains no-span. The bare <code>stage02_convex_pose_velocity_0p00_z</code> row spans locally at 1.327×10^{-15} , confirming the endpoint limit of that family, but this is not an accepted full-TFE replacement. The bounded nonlinear smoke closes terminal velocity to 8.124×10^{-17} for that row but gives only 4.142/2.305 position/velocity order; the nonterminal <code>0p01</code> row leaves terminal velocity at 6.255×10^{-6} . Ultra-near linear extrapolation and three-point quadratic non-terminal extrapolation both close terminal velocity without projection, but both retain the same 4.142/2.305 order loss. The diagnostic ledger also retains the missing-direction decomposition and source-to-velocity evidence: the 34-row audit based on the current-pose endpoint velocity matrix does not span; the matrix-difference branch records 0.042593 before scale refinement, 0.009527 after scale refinement, and 0.009590 for the normalized-history source law. The next repair must therefore derive a nonterminal endpoint-pose predictor that closes terminal velocity on trajectory without order loss.
Sparse backend speed		Open engineering caveat. Row-VJP reduces seed count but is not yet a stable runtime win over dense <code>jacfwd</code> .
Sharp-friction regime	coarse	Open practical caveat. Deeper fixed refinement recovers high order, but the coarse Brown–McPhee sharp case remains order-reduced and more expensive.